

CONFLUENT CLOUD · REAL-TIME FRAUD DETECTION

Scam Shield

A screenshot walkthrough of the live platform — payment events to a governed fraud decision, the Terraform that builds it, and the agentic ops layer on top.

PII-Safe

Deterministic Core

Governed AI

IaC + Lakehouse

Confluent Kafka · Confluent Flink SQL · Schema Registry · Tableflow → Apache Iceberg · HashiCorp Terraform · on Microsoft Azure (East US)

How to read this walkthrough

Each slide pairs a real console screenshot with a short, self-contained explanation — no voiceover needed. The deck self-runs and exports straight to video.

01

Agentic Ops Dashboard

Command center, investigation, audit, learning loop.

02

Terraform IaC

Two roots in Git; plan/apply through GitHub Actions.

03

Platform & Backbone

One Confluent account, one Kafka cluster on Azure.

04

Stream Processing

Eight Flink SQL statement sets are the detection core.

05

Connectors & Reliability

Managed Postgres sinks; dead-letter queues.

06

Topics & Schemas

32 governed topics, each an Avro data contract.

07

Governed AI Narration

ML_PREDICT writes analyst notes inside Flink.

08

Cost & Governance

Advanced governance; itemised, credit-offset billing.

SECTION · 01 · GOVERNED AI & AGENTIC OPS

Agentic Ops Dashboard

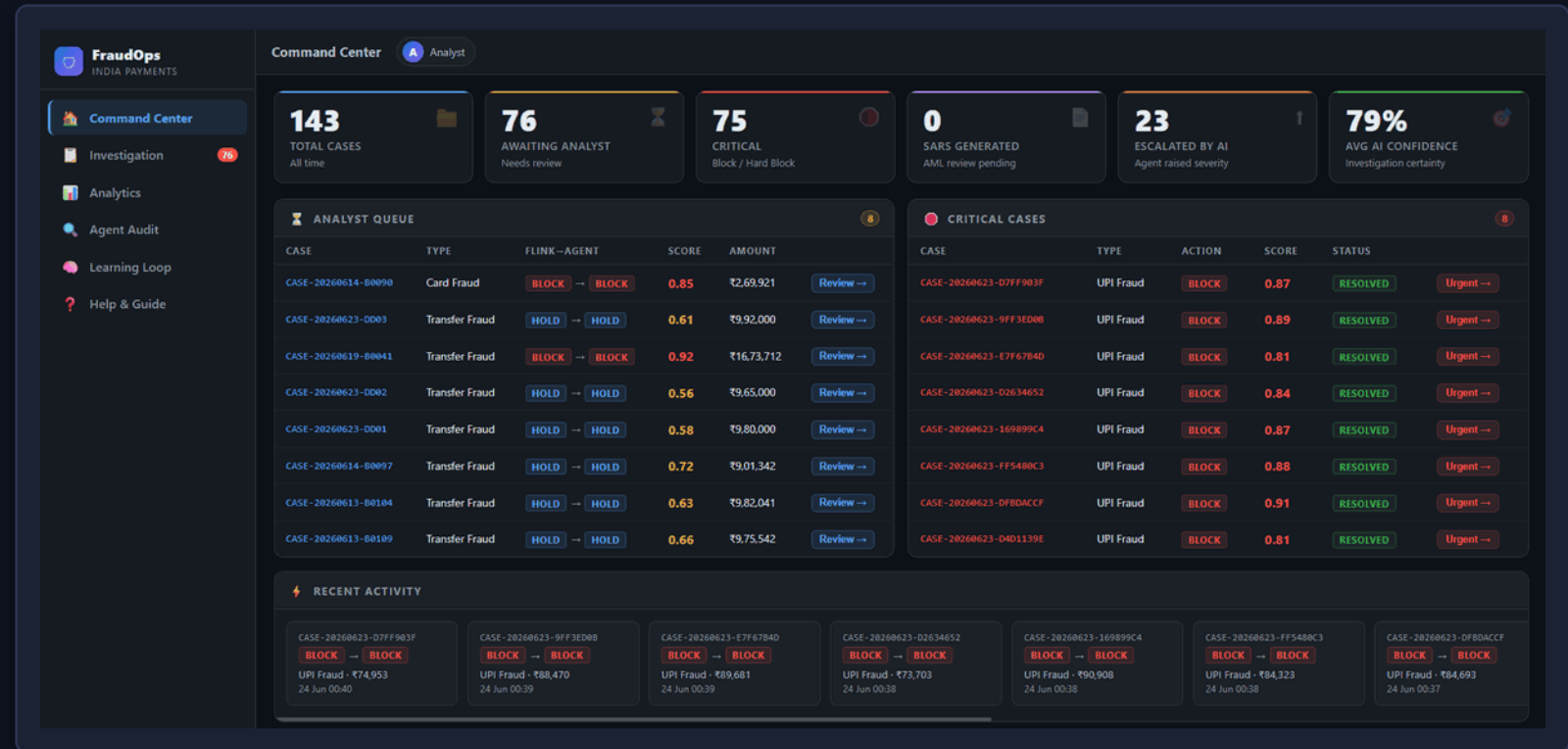
A governed agentic layer triages, investigates, audits and learns on top of the platform — every model call routed through the governance gateway.

A live command center for fraud ops

WHAT YOU'RE SEEING

- Real-time tiles: total, awaiting analyst, critical, escalated
- 79% of cases auto-handled; the rest queued for a human
- Analyst queue ranked by risk, action and SLA
- Fed by fraud.alert.raised + platform.ops.event over SSE

WHY IT MATTERS Analysts see the whole fraud picture and act where it matters most.



Deterministic score, agent cross-check

WHAT YOU'RE SEEING

- Flink rule score and agent confidence shown side by side
- Decision severity path from ALLOW through HARD_BLOCK
- Detected fraud patterns and per-rule validation listed
- Python tools cross-validate Flink hits — never re-score

WHY IT MATTERS *The agent corroborates the deterministic decision; it doesn't replace it (INV-3).*

FraudOps
INDIA PAYMENTS

Investigation Workspace

143 Total 76 Awaiting 75 Critical 0 SAR 23 Escalated 67 Resolved Analyst

CASES 143

Search case ID, bn, alert...

Block Hold Challenge Escalated SAR Awaiting

Risk Score Meters

FLINK SCORE 87%

AGENT CONFIDENCE 92%

RISK LEVEL CRITICAL

Decision Severity Path

ALLOW CHALLENGE HOLD BLOCK F+ A HARD BLOCK

Flink decision Agent recommendation Both agreed

Fraud Patterns Detected

SIM SWAP TAKEOVER

Rule Validation

SIM-SWAP-001
UNCONFIRMED N/A
No additional detail

APP-002
UNCONFIRMED N/A
No additional detail

APP-001-TFR
UNCONFIRMED N/A
No additional detail

MULE-001
UNCONFIRMED N/A
No additional detail

VEL-001
✓ CONFIRMED CRITICAL
No additional detail

Investigation Narrative

Investigation Findings

Analyst Narrative for UPI Transaction TXN-UPI-20260623-P0010

1. Rules Fired and Their Significance - VEL-001 (CRITICAL): This rule fired because the transaction amount (Rs74,953) is 749.53 times the account's 30-day average, exceeding the >100x threshold. This extreme deviation from the account's typical behavior is a strong indicator of potential fraudulent activity, as legitimate transactions rarely deviate so drastically from established patterns. The rule's confirmation and critical severity underscore the need for immediate action to prevent potential financial loss and account compromise.

Every alert gets a plain-English narrative

WHAT YOU'RE SEEING

- GPT-4o drafts a structured investigation narrative per case
- It cites the specific rules, signals and recommended action
- All model calls routed through the AI Governance gateway
- PII-free — the AI never sees raw identifiers (INV-1)

WHY IT MATTERS *Investigators read why, not just a score — fully governed and auditable.*

FraudOps
INDIA PAYMENTS

Investigation Workspace

143 Total 76 Awaiting 75 Critical 0 SAR 23 Escalated 67 Resolved Analyst

CASES 143

Search case ID, bn, alert...

Block Hold Challenge Escalated

SAR Awaiting

Block → BLOCK CRITICAL
CASE-20260623-07FF903F UPI Fraud 0.87
24 Jun 00:40

Block → BLOCK CRITICAL
CASE-20260623-9FF3ED00 UPI Fraud 0.89
24 Jun 00:39

Block → BLOCK CRITICAL
CASE-20260623-E7F67B40 UPI Fraud 0.81
24 Jun 00:39

Block → BLOCK CRITICAL
CASE-20260623-02634652 UPI Fraud 0.84
24 Jun 00:38

Block → BLOCK CRITICAL
CASE-20260623-169899C4 UPI Fraud 0.87
24 Jun 00:38

Block → BLOCK CRITICAL
CASE-20260623-FF5480C3 UPI Fraud 0.88
24 Jun 00:38

Block → BLOCK CRITICAL
CASE-20260623-DFB0ACCF UPI Fraud 0.91
24 Jun 00:37

Block → BLOCK CRITICAL
CASE-20260623-0401139E UPI Fraud 0.81
24 Jun 00:37

Block → BLOCK CRITICAL
CASE-20260623-2B7F06FF UPI Fraud 0.83
24 Jun 00:36

Block → BLOCK CRITICAL
CASE-20260623-434221A9 UPI Fraud 0.88
24 Jun 00:36

Investigation Narrative

Investigation Findings

Analyst Narrative for UPI Transaction TXN-UPI-20260623-P0010

1. Rules Fired and Their Significance - VEL-001 (CRITICAL): This rule fired because the transaction amount (Rs74,953) is 749.53 times the account's 30-day average, exceeding the >100x threshold. This extreme deviation from the account's typical behavior is a strong indicator of potential fraudulent activity, as legitimate transactions rarely deviate so drastically from established patterns. The rule's confirmation and critical severity underscore the need for immediate action to prevent potential financial loss and account compromise.

2. Behavioral Signals - Supporting Signals: - The high behavioral risk score aligns with the VEL-001 rule's findings, suggesting a potential SIM swap takeover attempt. This pattern is consistent with unauthorized access attempts where an attacker intercepts the SIM card to gain control over the UPI account. - The absence of SIM change detection (SIM-SWAP-001) and the lack of new device or first-time beneficiary flags (APP-002) indicate that the account's usual security parameters have not been altered, which is unusual for legitimate transactions of this magnitude. - Contradictory Signals: - The MuleHunter score (0.180) is below the mule threshold (>0.75), suggesting that the account does not exhibit typical mule account behaviors. This lack of corroboration from the MuleHunter model weakens the suspicion of money muling activities, focusing the investigation primarily on the SIM swap risk.

3. Recommended Action and Confidence Level - Recommended Action: Initiate a multi-factor authentication (MFA) challenge for the account holder to verify their identity. This could include sending a one-time password (OTP) to the registered mobile number or email, or prompting the account holder to answer security questions. Additionally, monitor the account for any further suspicious activity and consider temporarily restricting high-value transactions until the account's security can be confirmed. - Confidence Level: 0.92. The high confidence level is based on the strong evidence from the VEL-001 rule, which indicates a significant deviation from the account's normal behavior. The behavioral risk score further supports the suspicion of a SIM swap takeover attempt. While the MuleHunter score does not corroborate money muling activities, the overall pattern of evidence strongly suggests that immediate action is necessary to secure the account and prevent potential fraud.

Additional Context

Investigation Findings

The transaction was flagged because it triggered two high-risk rules: 'SIM-SWAP-001' and 'VEL-001', which are associated with the 'SIM_SWAP_UPI_TAKEOVER' pattern. The behavioral risk score is also high, indicating a potential SIM swap takeover attempt, where an attacker may have gained unauthorized access to the customer's UPI account by intercepting the SIM card.

Live · 10:23

Outcomes, trends and model impact

WHAT YOU'RE SEEING

- Detection rate, false-positive and auto-handled KPIs
- Geographic and time-of-day fraud distribution
- Live fraud intelligence by channel and typology
- AI escalation impact: agent-handled vs analyst-confirmed

WHY IT MATTERS Leadership sees the platform's effectiveness and where to tune it.



Every agent action is on the record

WHAT YOU'RE SEEING

- Append-only audit trail of every tool call and decision
- Each row ties to a case, actor, action and Kafka event
- An on-screen assistant answers questions over live cases
- Chat is RAG over current cases + the Qdrant knowledge base

WHY IT MATTERS *Full traceability plus an analyst copilot, grounded in real case data.*

Command Center

Investigation

Analytics

Agent Audit

Learning Loop

Help & Guide

Agent Audit Trail

143 Total
76 Awaiting
75 Critical
0 SAR
23 Escalated
67 Resolved
Analyst

#	TIMESTAMP	CASE ID	ACTOR	ACTION	DETAIL
1	24 Jun 00:40	CASE-20260623-D7FF903F	output_node	investigation_complete	confidence=0.92 score=0.87 ops_event=False
2	24 Jun 00:40	CASE-20260623-D7FF903F	knowledge_update_node	knowledge_updated	clarity=CLEAR_FRAUD doc_id=None learning=Extreme
3	24 Jun 00:40	CASE-20260623-D7FF903F	compliance_node	compliance_check	ncrp_filed=False mule_reported=False action=block ty...
4	24 Jun 00:40	CASE-20260623-D7FF903F	notification_node	notification_dispatched	action=freeze_account sent=False channel=upi
5	24 Jun 00:40	CASE-20260623-D7FF903F	case_management	case_opened	type=FraudCase
6	24 Jun 00:40	ALERT-20260623-P0010	decision_node	hitt_decision	hitt=False auto_action=freeze_account reason=BLOCK...
7	24 Jun 00:40	ALERT-20260623-P0010	upi_fraud_node	upi_specialist_analysis	confirmed_rules=1/5 confidence=0.92 recommendatio...
8	24 Jun 00:39	ALERT-20260623-P0010	context_enrichment_node	context_enriched	risk=HIGH typologies=[SIM_SWAP_TAKEOVER] runbo...
9	24 Jun 00:39	ALERT-20260623-P0010	orchestrator_node	alert_received	channel=UPI action=block score=0.87
10	24 Jun 00:39	CASE-20260623-9FF3ED08	output_node	investigation_complete	
11	24 Jun 00:39	CASE-20260623-9FF3ED08	knowledge_update_node	knowledge_updated	
12	24 Jun 00:39	CASE-20260623-9FF3ED08	compliance_node	compliance_check	
13	24 Jun 00:39	CASE-20260623-9FF3ED08	notification_node	notification_dispatched	
14	24 Jun 00:39	CASE-20260623-9FF3ED08	case_management	case_opened	
15	24 Jun 00:39	ALERT-20260623-P0009	decision_node	hitt_decision	
16	24 Jun 00:39	ALERT-20260623-P0009	upi_fraud_node	upi_specialist_analysis	
17	24 Jun 00:39	ALERT-20260623-P0009	context_enrichment_node	context_enriched	
18	24 Jun 00:39	ALERT-20260623-P0009	orchestrator_node	alert_received	
19	24 Jun 00:39	CASE-20260623-E7F67B4D	output_node	investigation_complete	
20	24 Jun 00:39	CASE-20260623-E7F67B4D	knowledge_update_node	knowledge_updated	
21	24 Jun 00:39	CASE-20260623-E7F67B4D	compliance_node	compliance_check	
22	24 Jun 00:39	CASE-20260623-E7F67B4D	notification_node	notification_dispatched	
23	24 Jun 00:39	CASE-20260623-E7F67B4D	case_management	case_opened	
24	24 Jun 00:39	ALERT-20260623-P0008	decision_node	hitt_decision	hitt=False auto_action=freeze_account reason=Bl...
25	24 Jun 00:39	ALERT-20260623-P0008	upi_fraud_node	upi_specialist_analysis	confirmed_rules=1/5 confidence=0.86 recommend...

AI Analyst Assistant

help me with ALERT-20260623-P0010

ASSISTANT

ALERT-20260623-P0010 is a **blocked** transaction (action=block) that was automatically stopped by the system (hitt=False). It is not linked to any fraud case, SAR, or pattern cluster in the current data set. No further analyst action is required unless you have additional context.

YOU

What was the agent action?

ASSISTANT

The agent action for ALERT-20260623-P0010 was **BLOCK**.

Ask about cases, decisions, SARs...

Send

The platform proposes its own new rules

WHAT YOU'RE SEEING

- Agents cluster resolved cases into emerging fraud patterns
- Each gap becomes a PENDING Flink SQL rule proposal
- An analyst must approve before anything deploys
- Approved rules ship as code via a pull request

WHY IT MATTERS *The system learns and suggests — humans still approve every rule change (INV-3).*

FraudOps
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Command Center
Investigation 76
Analytics
Agent Audit
Learning Loop
Help & Guide

Learning Loop
AI-generated rule proposals and emerging fraud pattern clusters. Approve proposals to deploy new Flink rules.

RULE PROPOSALS 0 [Refresh](#)

No proposals yet — the agent will surface them as it processes cases.

EMERGING PATTERNS 10 [Run Now](#)

UPI < ₹1,000 per txn **10 cases**

SIM-Port Account Takeover
10 accounts were compromised following a SIM port attack. Once the attacker controlled the mobile number, they bypassed OTP authentication across banking apps and made transfers (under ₹1,000 each). All 10 cases triggered the same set of detection rules (bc7d3b), suggesting a common attack pattern.
bc7d3b
First seen: 2026-06-23 19:06 Latest: 2026-06-23 19:10 728756d1 [Investigate 10 cases →](#)

CARD ₹1,000 – ₹10,000 per txn **NOVEL** **4 cases**

Novel / Unclassified Pattern
4 CARD transactions (₹1,000–₹10,000 each) share an identical pattern but don't match any known fraud type in the system's rule library. No existing detection rule matches this signature (334c4a) triggered but produced no typology match). This could represent a new fraud method. Review the cases for common attributes and consider raising a rule proposal if the pattern holds.
334c4a
First seen: 2026-06-12 00:58 Latest: 2026-06-20 08:58

ACCOUNT TAKEOVER UPI
15 UPI cases showing account takeover upi behaviour triggered the same set of detection rules (cf1082), suggesting a common attack pattern.
cf1082
First seen: 2026-06-08 21:58 Latest: 2026-06-21 11:58

Novel pattern — Novel / Unclassified Pattern
4 CARD transactions (₹1,000–₹10,000 each) share an identical pattern but don't match any known fraud type in the system's rule library. No existing detection rule matches this signature (334c4a) triggered but produced no typology match). This could represent a new fraud method. Review the cases for common attributes and consider raising a rule proposal if the pattern holds.

Payment channel	CARD
Transaction amount range	₹1,000 – ₹10,000 per txn
Cases in this group	4 cases
Detection rules triggered	334c4a
First case seen	2026-06-12 00:58
Most recent case	2026-06-20 08:58

Recommended action: Flag for analyst review. Run pattern gap detection. Consider new Flink rule proposal.

TRANSFER ₹1,000 – ₹10,000 per txn **15 cases**

Transaction Structuring (Below CTR)
15 TRANSFER transactions were deliberately sized to stay just below the ₹10 lakh reporting threshold (₹1,000–₹10,000 per txn).
First seen: 2026-06-08 21:58 Latest: 2026-06-21 11:58

Live · 10:26

SECTION · 02 · INFRASTRUCTURE AS CODE

Terraform Platform

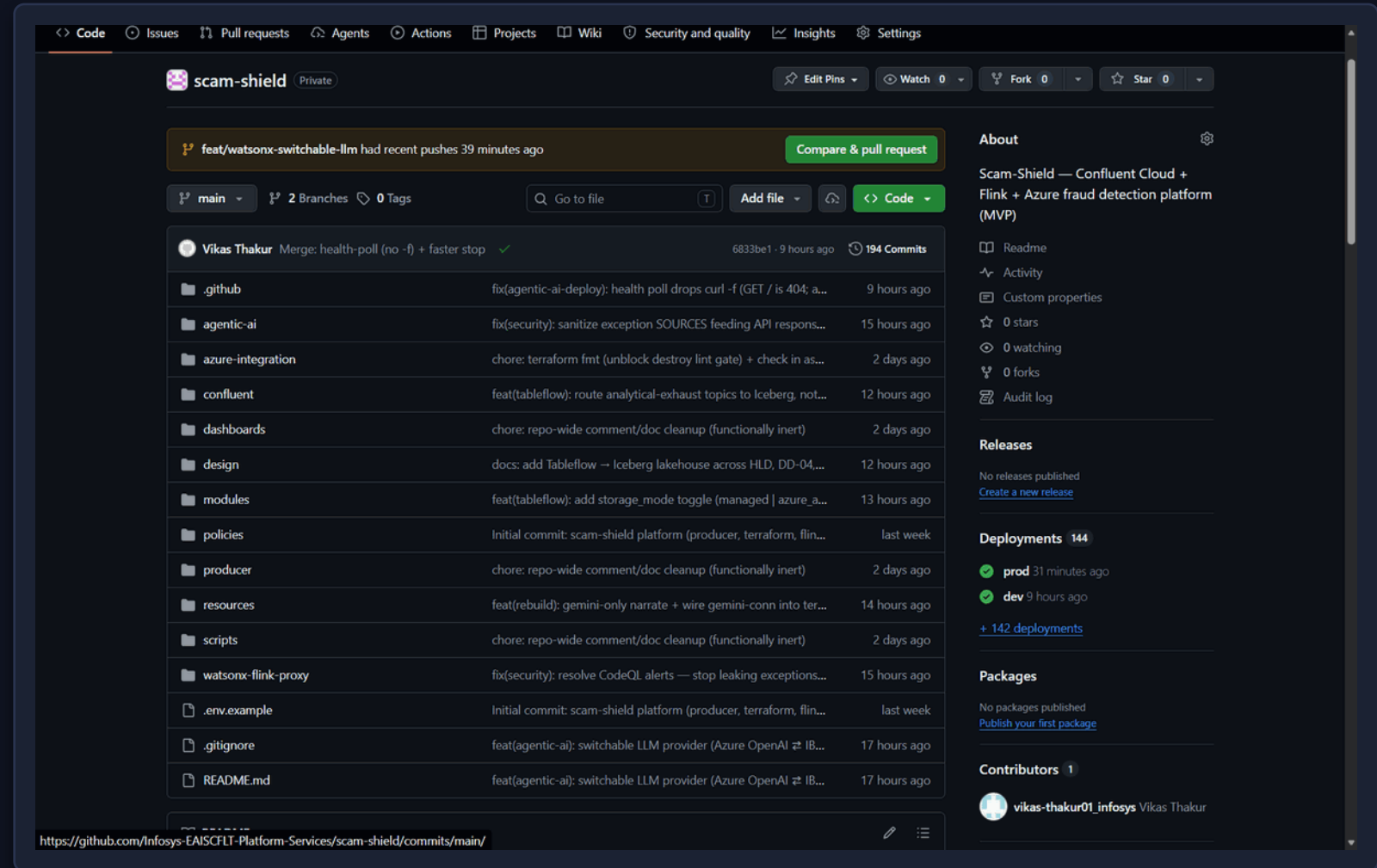
Two Terraform roots in one Git repo provision every Confluent and Azure resource — planned, reviewed and applied through GitHub Actions.

Everything is code, in one repo

WHAT YOU'RE SEEING

- One repo holds both roots: confluent/ and azure-integration/
- modules/, resources/ and policies/ sit beside the app code
- No resource is ever created by hand in a console
- Drift is reconciled back to Git on every run

WHY IT MATTERS *Terraform is the sole source of truth — if it's not in Git, it doesn't exist (TF-1).*

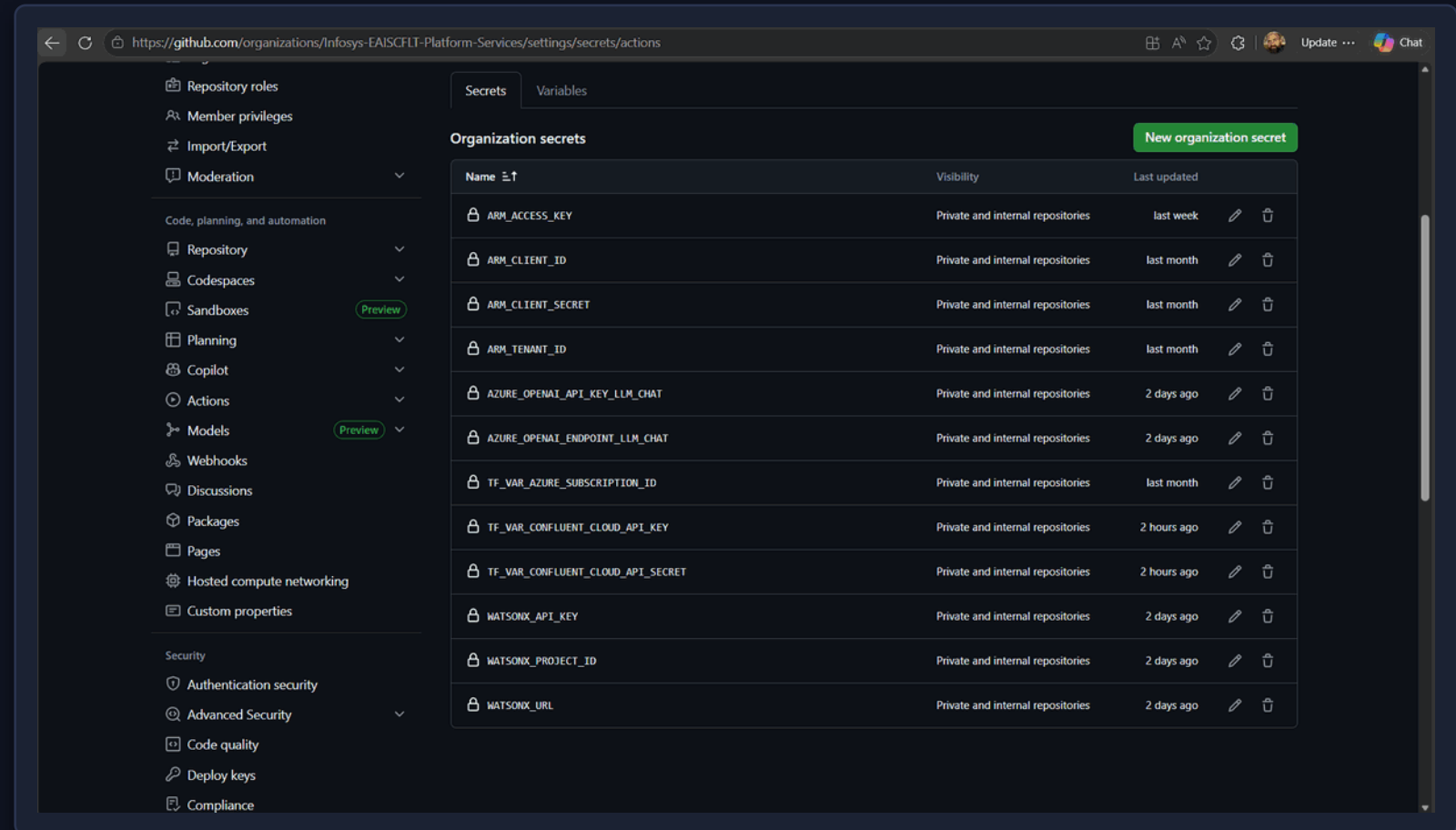


Config is committed; secrets never are

WHAT YOU'RE SEEING

- tfvars and the lock file are committed for reproducibility
- API keys & connection secrets live only in GitHub Secrets
- Injected at run time via `TF_VAR_*` — never written to disk
- Secret variables are marked `sensitive = true`

WHY IT MATTERS A clean split of committed config from injected secrets (TF-2).



Lint → plan → approve → apply

WHAT YOU'RE SEEING

- Every change flows through a GitHub Actions pipeline
- Format-check and validate gate each commit
- Plans are posted to the PR for review before apply
- prod / prod-dr applies require mandatory approval

WHY IT MATTERS *No one applies from a laptop — promotion is a reviewed, auditable pipeline.*

The screenshot displays the GitHub Actions interface for the repository 'Infosys-EAISCFLT-Platform-Services' under the branch 'scam-shield'. The 'Actions' tab is selected, showing a list of workflow runs. A notification banner at the top indicates an upcoming change to the GitHub App installation token format. The 'All workflows' section shows 573 workflow runs. The table below lists the most recent runs:

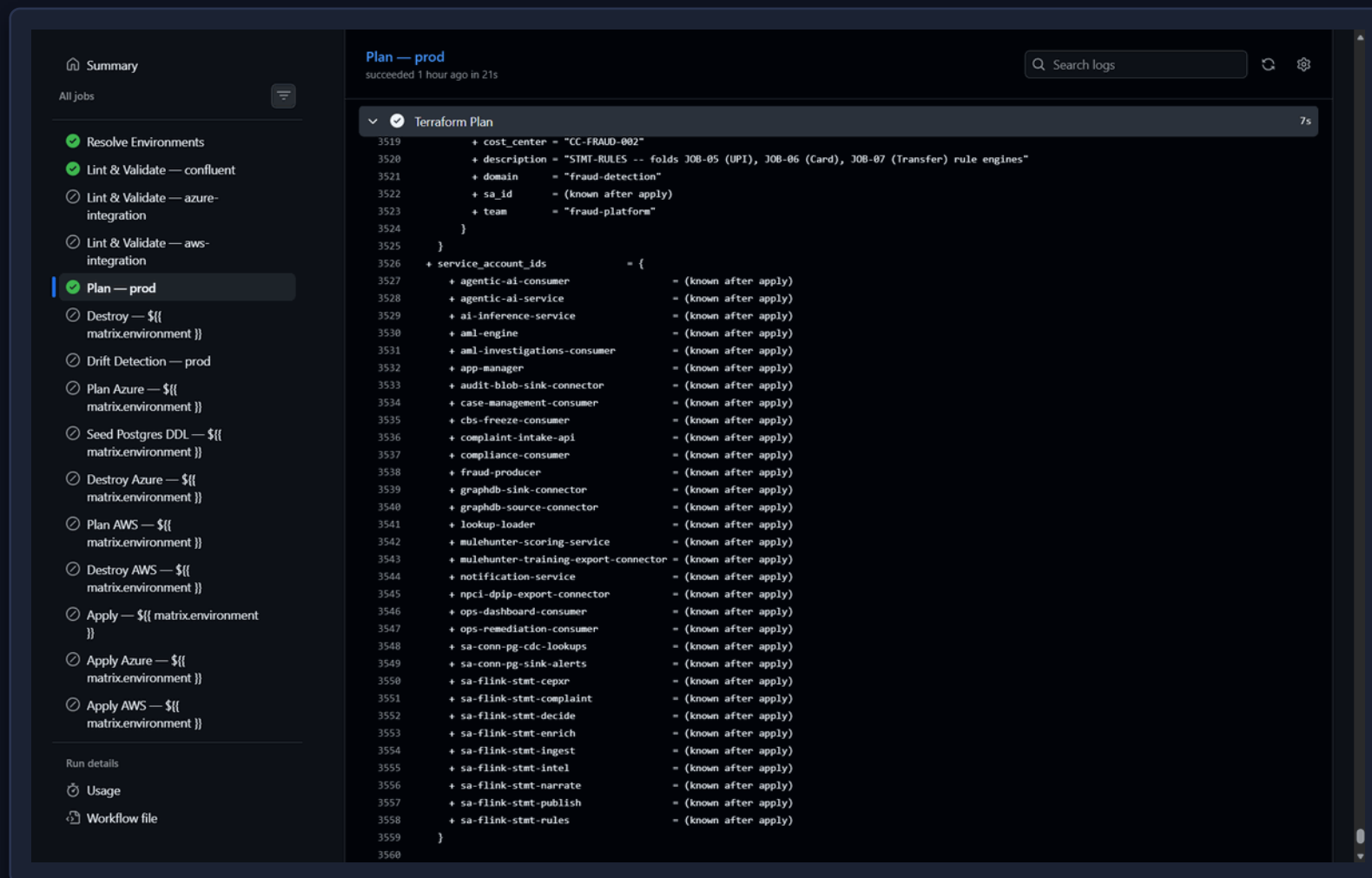
Workflow	Event	Status	Branch	Actor	Run ID	Run Name	Run Time	Run Date
Terraform — Confluent Cloud	Manually run by vikas-thakur01_infosys	Completed	feat/watsonx-switchable-11m	vikas-thakur01_infosys	#226	Terraform — Confluent Cloud #226: Manually run by vikas-thakur01_infosys	12m 51s	40 minutes ago
Terraform — Confluent Cloud	Manually run by vikas-thakur01_infosys	Completed	feat/watsonx-switchable-11m	vikas-thakur01_infosys	#225	Terraform — Confluent Cloud #225: Manually run by vikas-thakur01_infosys	50s	42 minutes ago
Terraform — Confluent Cloud	Manually run by vikas-thakur01_infosys	Completed	feat/watsonx-switchable-11m	vikas-thakur01_infosys	#224	Terraform — Confluent Cloud #224: Manually run by vikas-thakur01_infosys	50s	47 minutes ago
Terraform — Confluent Cloud	Manually run by vikas-thakur01_infosys	Completed	main	vikas-thakur01_infosys	#223	Terraform — Confluent Cloud #223: Manually run by vikas-thakur01_infosys	1m 18s	Today at 8:54 AM
Terraform — Confluent Cloud	Manually run by vikas-thakur01_infosys	Completed	main	vikas-thakur01_infosys	#222	Terraform — Confluent Cloud #222: Manually run by vikas-thakur01_infosys	46s	Today at 8:52 AM
Terraform — Confluent Cloud	Manually run by vikas-thakur01_infosys	Completed	main	vikas-thakur01_infosys	#221	Terraform — Confluent Cloud #221: Manually run by vikas-thakur01_infosys	54s	Today at 8:32 AM
Agentic AI — Build & Deploy to Jumphost	Manually run by vikas-thakur01_infosys	Completed	main	vikas-thakur01_infosys	#10	Agentic AI — Build & Deploy to Jumphost #10: Manually run by vikas-thakur01_infosys	2m 34s	Today at 12:17 AM

Every change is a reviewed plan first

WHAT YOU'RE SEEING

- terraform plan runs per environment with its var-file
- Resources to add / change / destroy are shown explicitly
- The saved plan artifact is exactly what apply consumes
- Daily scheduled plans detect drift on prod

WHY IT MATTERS *Changes are previewed and approved before they ever touch the platform.*



One pipeline provisions Confluent and Azure

WHAT YOU'RE SEEING

- confluent/ applies first; azure-integration/ reads its state
- Key Vault, PrivateLink, Postgres and monitoring provisioned
- Apply consumes the approved plan with -auto-approve
- Outputs hand off cleanly between the two roots

WHY IT MATTERS *The whole platform — both clouds — stands up from code in one flow.*

The screenshot displays a GitHub Actions workflow run titled "Apply Azure — prod" which succeeded 25 minutes ago. The workflow is part of the "Terraform Apply — azure-integration" job. The output shows the execution of Terraform Apply, including a warning about incomplete changes and a list of outputs for various resources.

Summary

All jobs

- Resolve Environments
- Lint & Validate — confluent
- Lint & Validate — azure-integration
- Lint & Validate — aws-integration
- Plan — \${{ matrix.environment }}
- Destroy — \${{ matrix.environment }}
- Drift Detection — prod
- Plan Azure — prod
- Seed Postgres DDL — \${{ matrix.environment }}
- Destroy Azure — \${{ matrix.environment }}
- Plan AWS — \${{ matrix.environment }}
- Destroy AWS — \${{ matrix.environment }}
- Apply — \${{ matrix.environment }}
- Apply Azure — prod**
- Apply AWS — \${{ matrix.environment }}

Run details

- Usage
- Workflow file

Apply Azure — prod
succeeded 25 minutes ago in 11m 59s

Search logs

Terraform Apply — azure-integration 11m 47s

```

127 azure_private_dns_record_ccn_wildcard[0]: Creation complete after 1s [id=/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.Network/privateDnsZones/g0do07.eastus.azure.confluent.cloud/A/*]
128
129 Warning: Applied changes may be incomplete
130
131 The plan was created with the -target option in effect, so some changes
132 requested in the configuration may have been ignored and the output values
133 may not be fully updated. Run the following command to verify that no other
134 changes are pending:
135     terraform plan
136
137 Note that the -target option is not suitable for routine use, and is
138 provided only for exceptional situations such as recovering from errors or
139 mistakes, or when Terraform specifically suggests to use it as part of an
140 error message.
141
142
143 Apply complete! Resources: 7 added, 0 changed, 5 destroyed.
144
145 Outputs:
146
147 action_group_id = "/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.Insights/actionGroups/ag-confluent-prod"
148 ccn_private_endpoint_ip = "100.78.217.29"
149 confluent_network_dns_domain = "g0do07.eastus.azure.confluent.cloud"
150 confluent_network_id = "n-g0do07"
151 hackathon_postgres_admin_password = <sensitive>
152 hackathon_postgres_admin_username = "scamadmin"
153 hackathon_postgres_database_name = "scam_shield"
154 hackathon_postgres_fqdn = "psql-scamshield-prod-ppc.postgres.database.azure.com"
155 hackathon_postgres_server_id = "/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.DBforPostgreSQL/flexibleServers/psql-scamshield-prod-ppc"
156 log_analytics_workspace_id = "/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.OperationalInsights/workspaces/law-confluent-prod"
157 log_analytics_workspace_name = "law-confluent-prod"
158 resource_group_id = "/subscriptions/***/resourceGroups/rg-confluent-prod"
159 workbook_id = "/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.Insights/workbooks/b4bb9e69-98d1-5aa1-9b74-73afcb3ca6bc"
  
```

> Post Run actions/checkout@v4 0s

> Complete job 0s

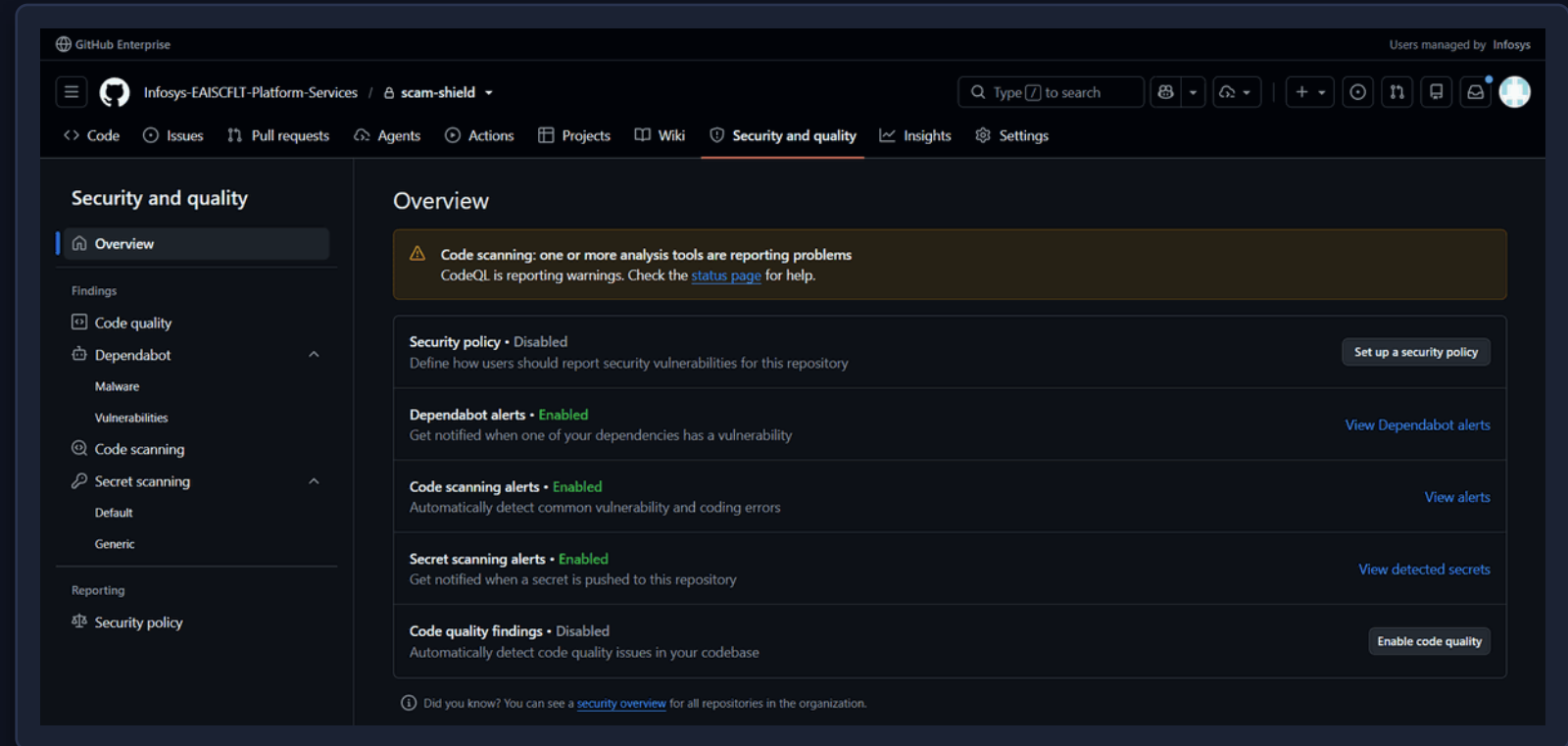
<https://github.com/Infosys-EAISFLT-Platform-Services/scam-shield/actions/runs/28073978849/job/83114296837>

Clean code, enforced automatically

WHAT YOU'RE SEEING

- Secret scanning blocks credentials from being committed
- Code scanning (CodeQL) flags vulnerabilities on push
- Dependabot tracks dependency risk
- OPA / Conftest gates block non-compliant prod plans

WHY IT MATTERS *Security and policy are enforced in CI, not left to a reviewer's memory.*



SECTION · 03 · THE LIVE PLATFORM

Confluent Kafka & Flink

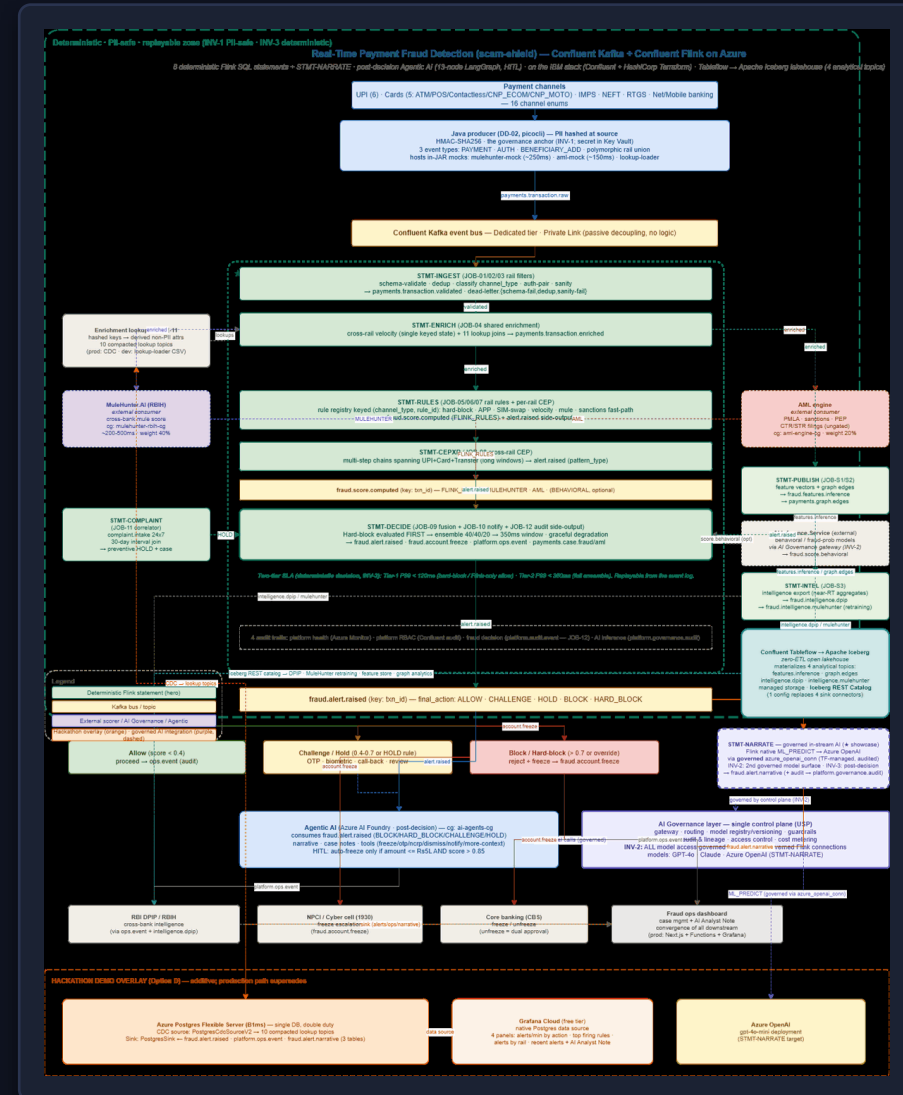
The running system the AI and Terraform sit on — payment events to a governed fraud decision, entirely on Confluent Kafka, Flink SQL and Schema Registry.

One Confluent account runs the whole platform

WHAT YOU'RE SEEING

- Production org: 1 environment, 1 Kafka cluster
- 32 Kafka topics carry every event end-to-end
- 3 managed connectors bridge Kafka ↔ PostgreSQL
- 1 Flink compute pool runs all stream processing

WHY IT MATTERS *Kafka is the single source of truth — transactions, enrichment, decisions and AI notes all flow through these topics.*

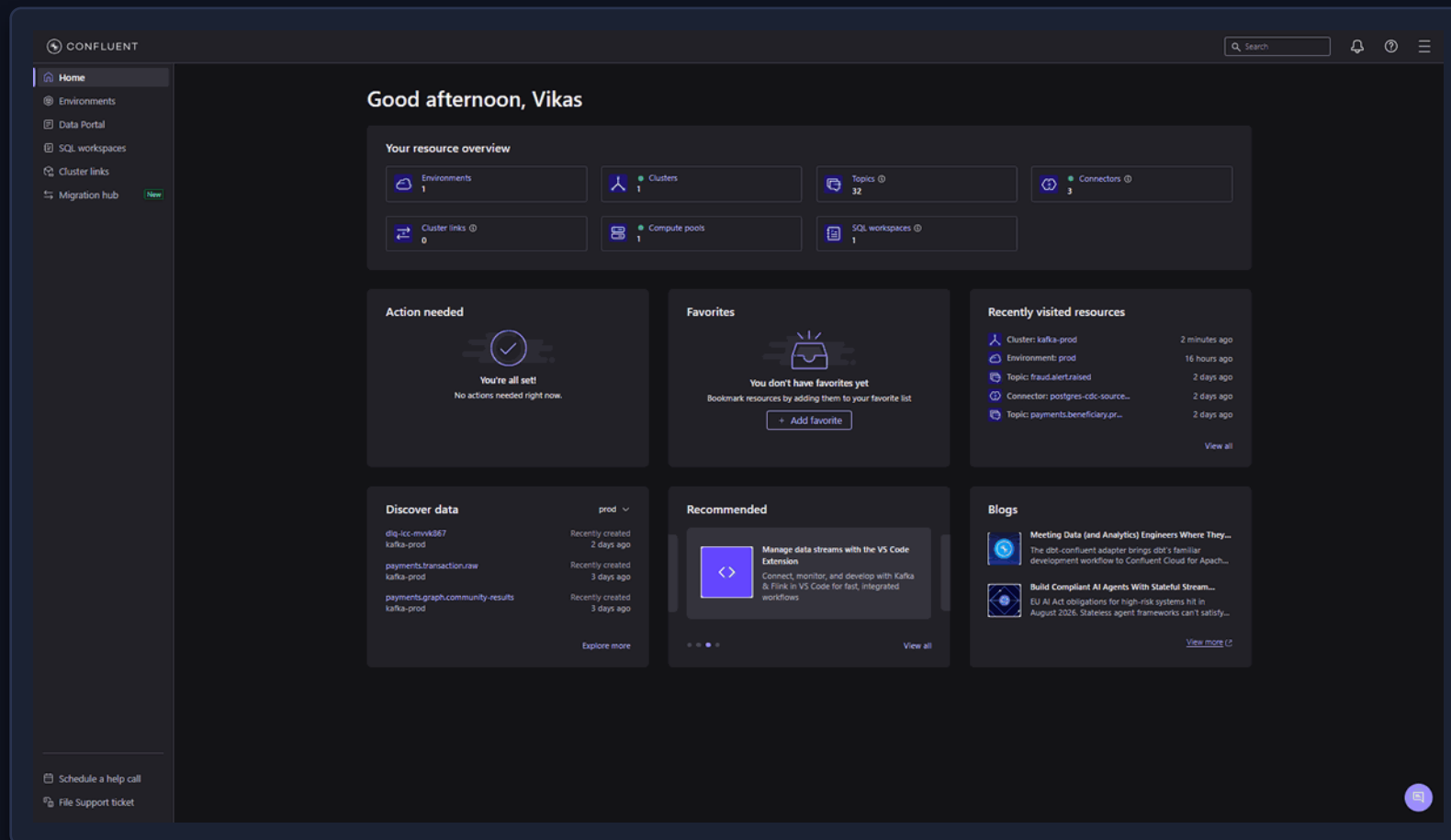


A dedicated, private cluster on Azure East US

WHAT YOU'RE SEEING

- Dedicated cluster on Private Link, status Running
- Provider Azure · region eastus · 274 partitions
- 32 topics, ~4.2 GB stored, live throughput shown
- Stream Lineage maps producers → topics → consumers

WHY IT MATTERS *Enterprise-grade isolation with an auditable, replayable event backbone.*

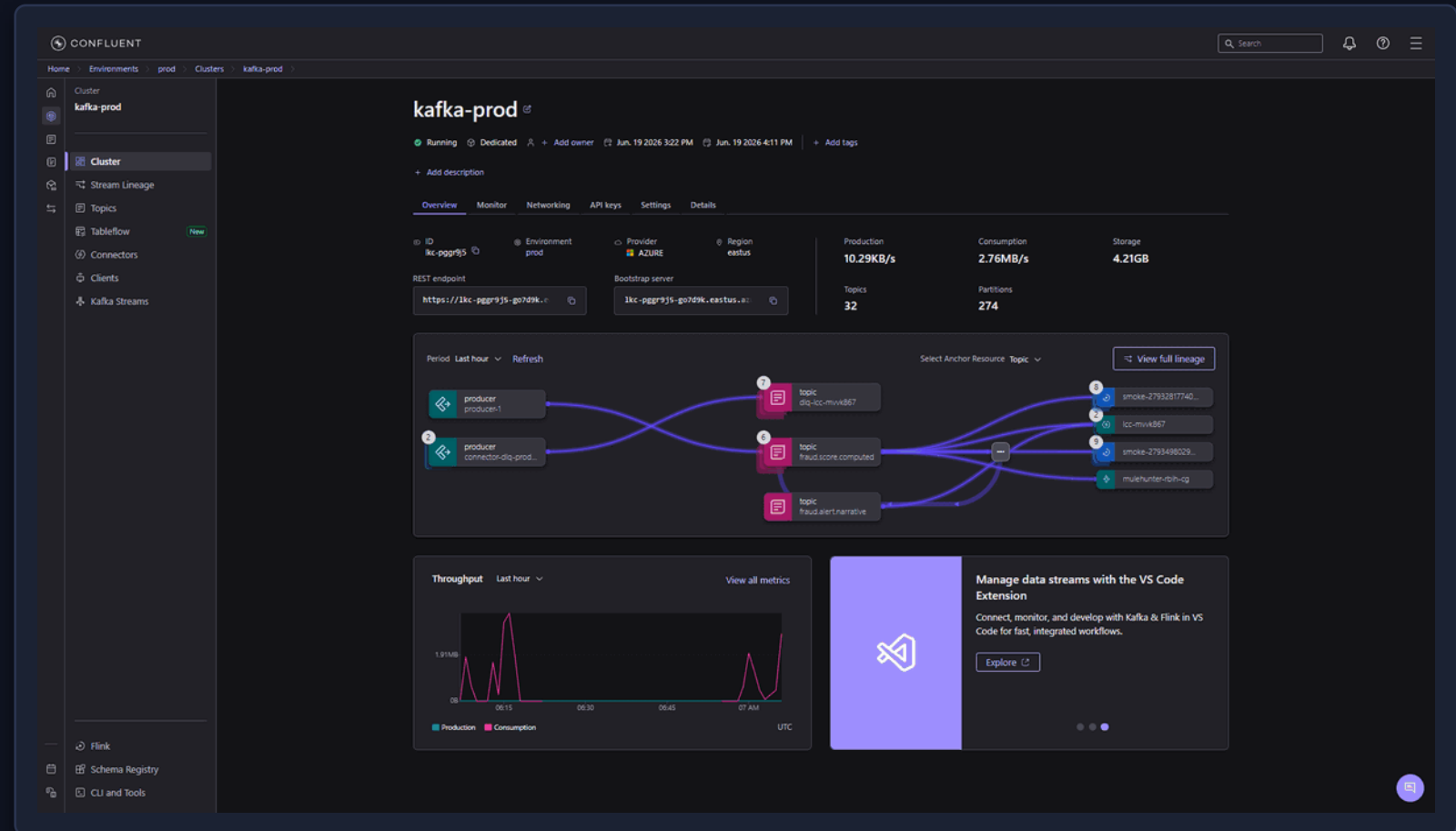


Every decision fans out atomically from STMT-DECIDE

WHAT YOU'RE SEEING

- STMT-DECIDE emits audit, ops and alert events together
- fraud.alert.raised feeds the sink, intel and complaint jobs
- platform.audit.event is the immutable 1-year trail
- One Flink checkpoint guarantees cross-topic consistency

WHY IT MATTERS *One decision produces many consistent outputs — never a partial state.*

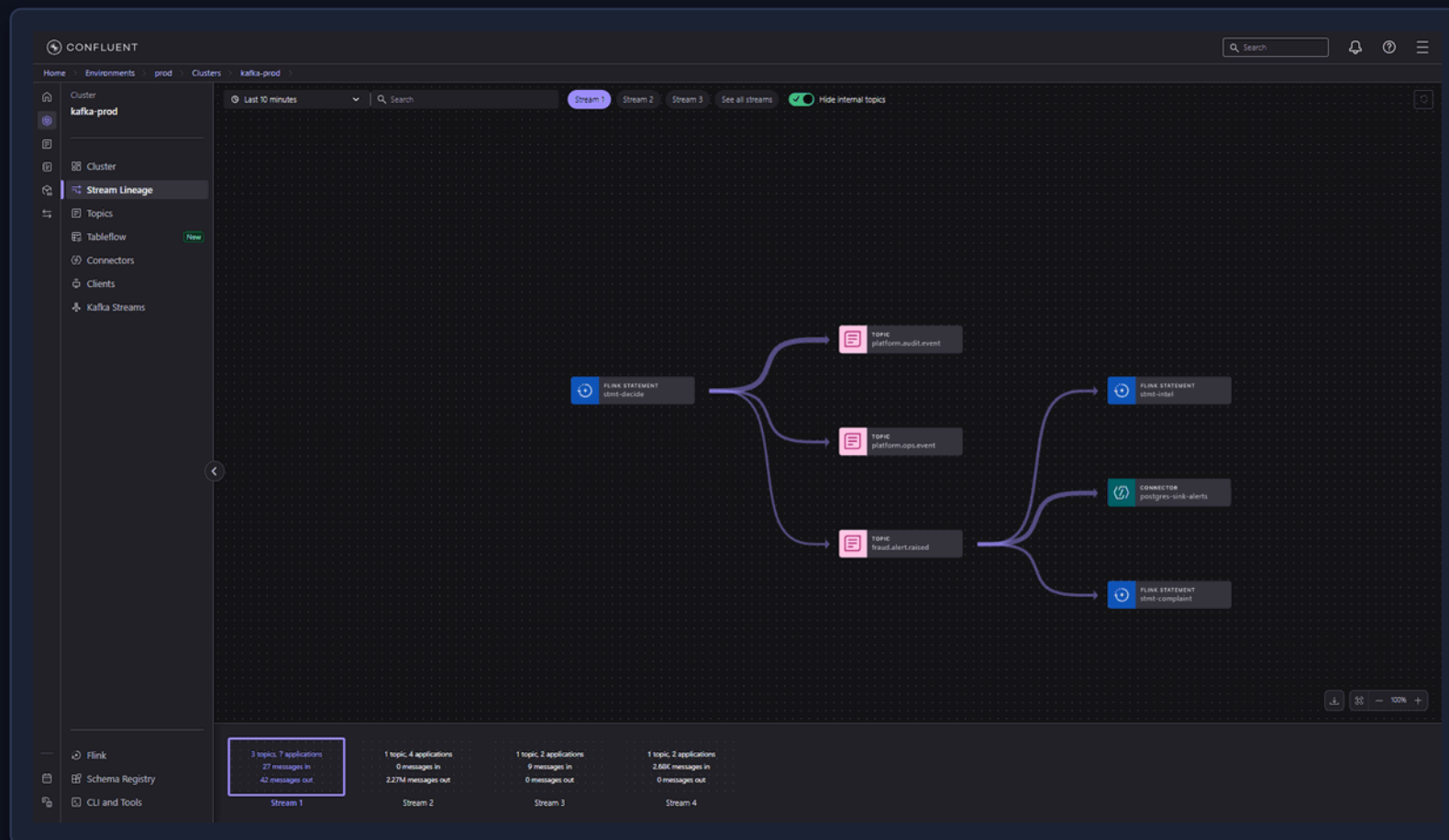


Regulatory intelligence is just Flink SQL

WHAT YOU'RE SEEING

- stmt-intel runs as an EXECUTE STATEMENT SET
- Reads fraud.alert.raised in 1-hour tumbling windows
- Writes the NPCI DPIP export + MuleHunter retraining feed
- Full SQL, service account and lineage visible in-console

WHY IT MATTERS *The detection core is pure SQL — no custom JARs to build or operate.*



Fraud decisions land in PostgreSQL via a managed sink

WHAT YOU'RE SEEING

- postgres-sink-alerts connector, status Running
- Consumes fraud.alert.raised + fraud.alert.narrative
- UPSERT keyed by alert_id into the operational store
- Powers the live Fraud Ops dashboard & case management

WHY IT MATTERS Operational egress is a fully managed connector — zero bespoke code.

The screenshot displays the Confluent Cloud interface for a stream named 'stm-intel'. The stream is in a 'Running' state. The right-hand panel shows the 'SQL Content' tab, which contains a complex SQL query. The query is a CTE (Common Table Expression) that defines a 'stm-intel' table. It includes various joins and filters, such as 'fraud.alert.raised', 'fraud.alert.narrative', and 'fraud.alert.raised'. The query is designed to be used as a managed sink for the 'stm-intel' stream.

```

1  --
2  -- stm-intel -- Intelligence export (2026-03)
3  --
4  -- v5.2 contract-sync (2026-06-17): conformed to registered Aero schema;
5  -- validate on cluster.
6  -- v5.3 P2-24 (2026-06-21): REVERTED for this statement - the OPTIONS hint is
7  -- NOT parseable inside a
8  -- TUMBLE TYP's TABLE() argument on Confluent Cloud Flink ("Encountered
9  -- /*+*/. Both TYP sources
10 -- was the default scan startup mode (acceptable: hourly, off the hot
11 -- path). The hint remains valid
12 -- on plain FROM scans (stm_cop/enrich/rules/publish).
13 -- v5.3 P2-24 (2026-06-21): DPDP log now groups by
14 -- beneficiary_account_id_hash (unlocked: field present
15 -- in fraud.alert.raised via Mulehunter dual-account schema + STM-
16 -- RULES populate id).
17 -- entity_id fixed (UNKNOWN_ENTITY -> beneficiary_account_id_hash);
18 -- currency and
19 -- total_blacklist_amount now aggregated from live data; WHERE filters
20 -- MALL beneficiary rows.
21 -- v5.3 P2-21 (2026-06-21): DPDP flagged_patterns uses MAX(pattern_type);
22 -- Mulehunter fund_flow_chain
23 -- populated from community-results fund_flow_chain when non-null.
24 -- v5.4 P2-22 (2026-06-21): mule_candidates populated with one representative
25 -- MuleCandidate row per
26 -- community window (community_id as placeholder, MAX(risk_score) for
27 -- candidacy_score).
28 -- MAX(risk_score) with GROUP BY columns + aggregates; requires cluster
29 -- validation.
30 -- Per-account UNNESTing of number_account_hashes deferred to offline
31 -- enrichment.
32 -- Statement : stm-intel
33 -- Service Acct : sa-fink-stm-intel
34 -- Owner : fraud-platform-team@infoys.com
35 -- Criticality : high / SLA: gold
36 --
37 -- Reads : fraud.alert.raised (1h tumbling windows)
38 -- Writes: fraud.intelligence-results (community_detector output)
39 -- fraud.intelligence-dpdp (DPDP feed -- per
40 -- entity)
41 -- fraud.intelligence.mulehunter (Mulehunter retraining
42 -- export)
43 --
  
```

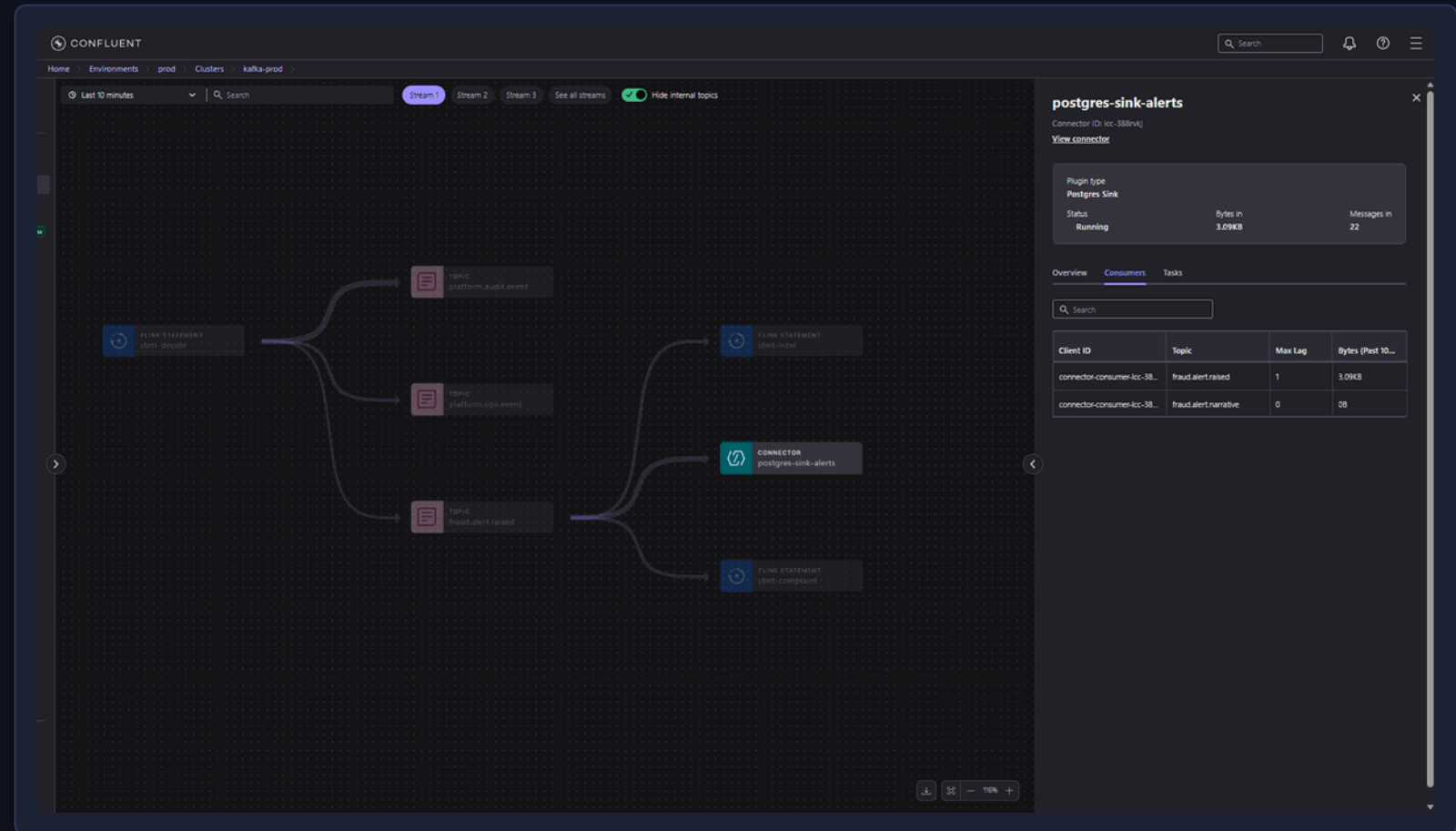
STREAM LINEAGE

A single score bus fuses every model

WHAT YOU'RE SEEING

- fraud.score.computed is the shared multi-model topic
- Flink rules, MuleHunter and the AML engine all write here
- STMT-DECIDE reads it in one 350 ms fusion window
- 40 / 40 / 20 weighted ensemble → one decision

WHY IT MATTERS *One topic, one window — simple, deterministic and replayable score fusion.*



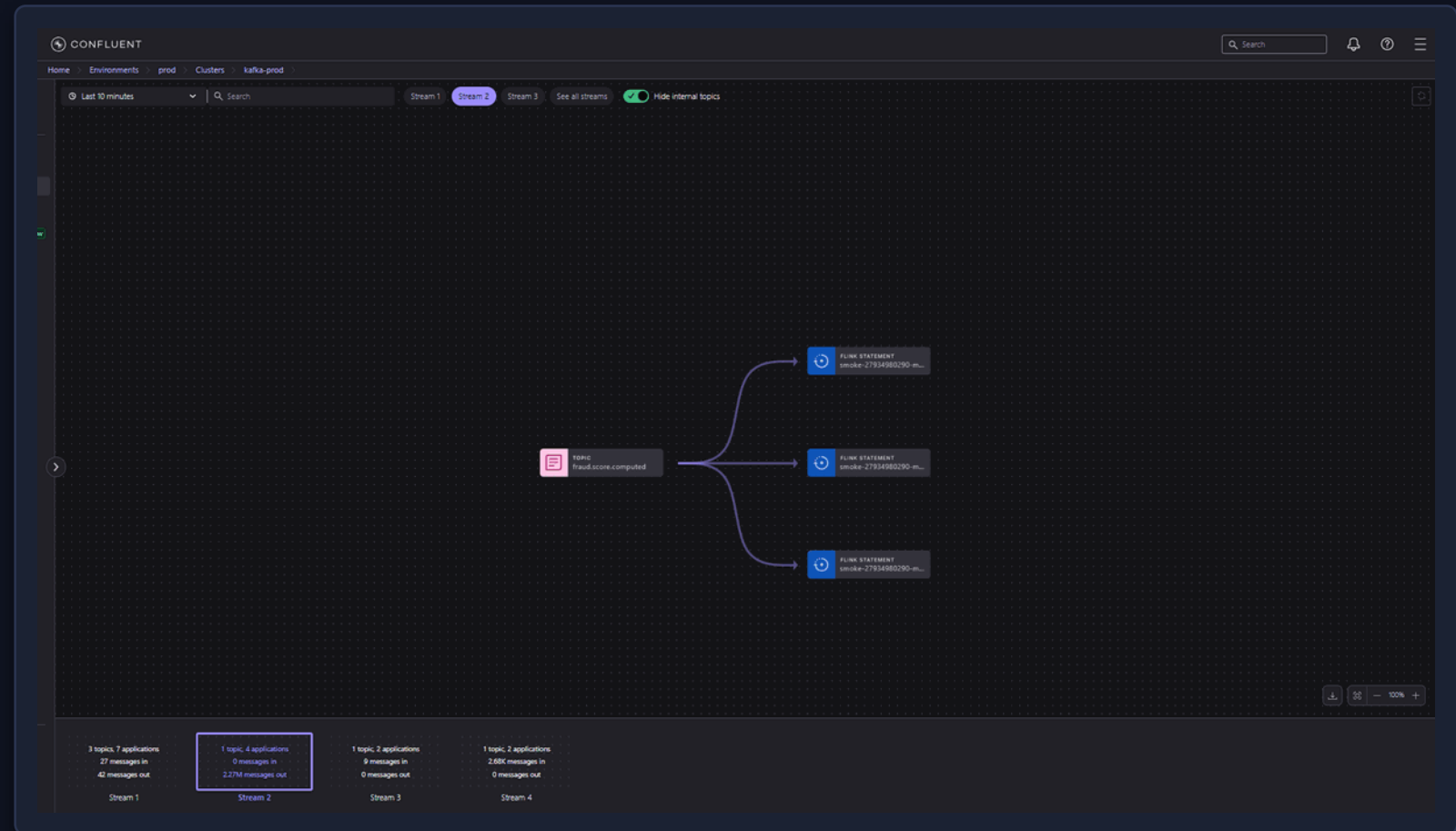
RELIABILITY

Bad events are quarantined, never dropped

WHAT YOU'RE SEEING

- A dead-letter producer captures schema & sanity failures
- Routed to dedicated payments.transaction.dead-letter topics
- Live byte and message metrics confirm healthy flow
- `errors.tolerance = all` keeps the pipeline running

WHY IT MATTERS *Malformed input is isolated for review instead of silently lost.*



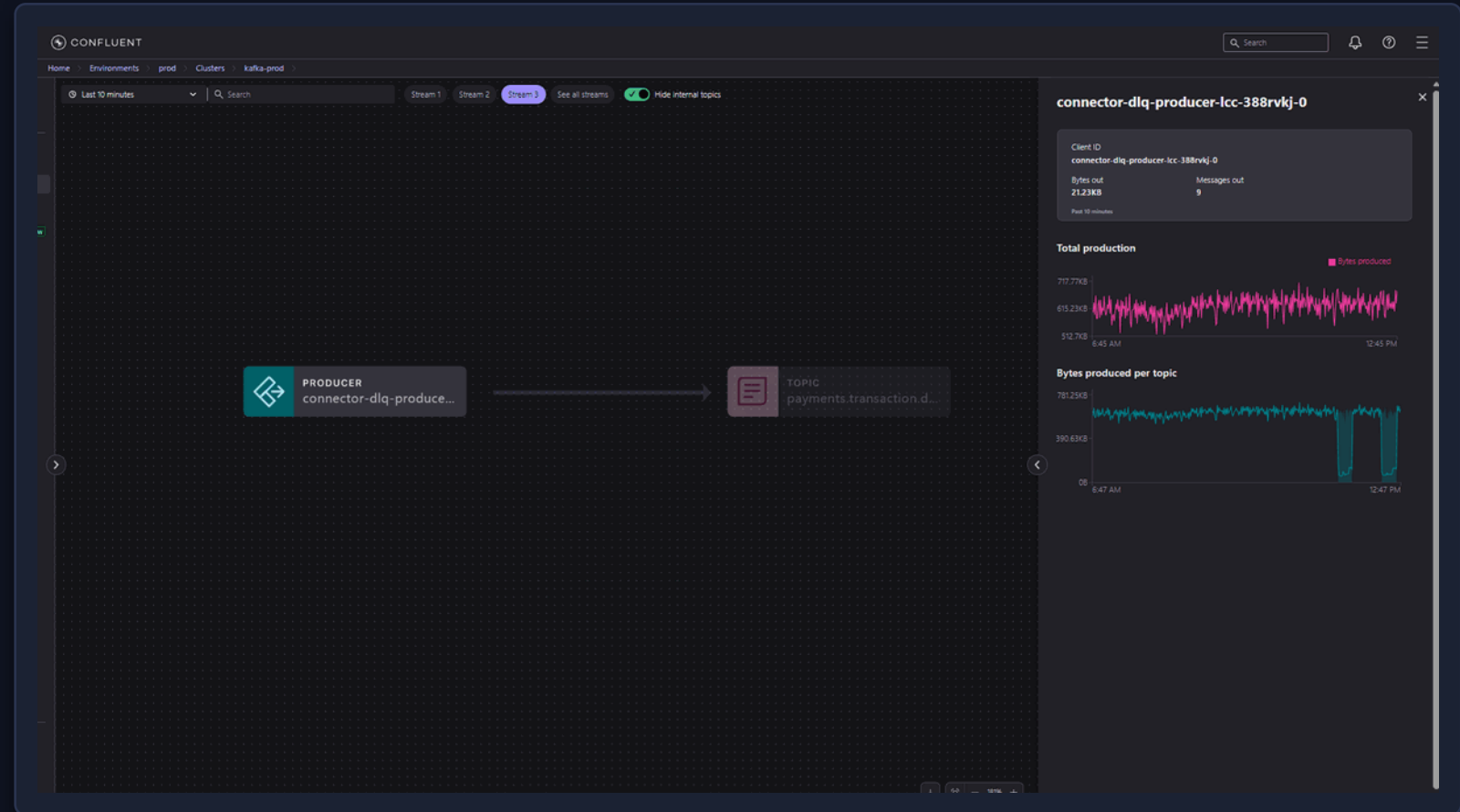
TOPICS

32 schema-governed topics, one per stage

WHAT YOU'RE SEEING

- Lookups, scores, alerts, intelligence and audit streams
- Network-protected — no ad-hoc edits over the internet
- Each topic carries a registered Avro data contract
- One-click 'Enable Tableflow' to the Iceberg lakehouse

WHY IT MATTERS *Every stage of the pipeline is an explicit, governed Kafka stream.*



Eight statement sets are the entire detection engine

WHAT YOU'RE SEEING

- ingest → enrich → rules → cep → decide → narrate
- Plus stmt-publish, stmt-intel and stmt-complaint
- All Running, 1 CFU each, zero messages behind
- Deployed and monitored on one shared compute pool

WHY IT MATTERS *The whole fraud core is declarative, observable Flink SQL.*

CONFLUENT

Home Environments prod Clusters kafka-prod

Cluster kafka-prod

Cluster Stream Lineage Topics Tableflow Connectors Clients Kafka Streams

Topics

Managing topics not available
Your browser can't create, delete, or change settings for topics of this cluster. All topics in this cluster are protected by a private network and cannot be altered over the Internet.
[Learn more \(i\) about managing these topics.](#)

Search topics Topics filter Show all CSV Add topic

Topic name	Tags	Partitions	Production	Consumption	Retained bytes	Consumers	Data contract	Tableflow sync st
compliance-sanctions-list	--	6	--	--	1.64KB	--	View data contract	Enable Tableflow
customers-account-profile	--	6	--	--	2.04KB	--	View data contract	Enable Tableflow
customers-device-profile	--	6	--	--	2.06KB	--	View data contract	Enable Tableflow
customers-geo-profile	--	6	--	--	2.02KB	--	View data contract	Enable Tableflow
customers-vulnerability-profile	--	6	--	--	1.65KB	--	View data contract	Enable Tableflow
dllo_kcc_movv8867	--	1	11.42KB/s	--	1.58KB	--	Add data contract	Enable Tableflow
fraud-account-freeze	--	6	--	--	1.03MB	--	View data contract	Enable Tableflow
fraud-alert-narrative	--	6	--	0B/s	0B	--	View data contract	Enable Tableflow
fraud-alert-raised	--	12	--	0B/s	2.07MB	--	View data contract	Enable Tableflow
fraud-alert-state	--	6	--	--	1.66KB	--	View data contract	Enable Tableflow
fraud-features-inference	--	12	--	--	691.21MB	--	View data contract	Enable Tableflow
fraud-intelligence-dep	--	6	--	--	25B	--	View data contract	Enable Tableflow
fraud-intelligence-malehunter	--	6	--	--	0B	--	View data contract	Enable Tableflow
fraud-score-behavioral	--	12	--	--	0B	--	View data contract	Enable Tableflow
fraud-score-computed	--	12	--	0B/s	178.24MB	--	View data contract	Enable Tableflow
fraud-velocity-counters	--	6	--	--	1.83KB	--	View data contract	Enable Tableflow

Flink Schema Registry CLI and Tools

Elastic compute that scales with load

WHAT YOU'RE SEEING

- flink-pool-prod-prod running on Azure eastus
- Using 8 of 30 CFUs — 27% of max capacity
- Live consumption and messages-in / out metrics
- Scales up under burst, back down when idle

WHY IT MATTERS *You pay for fraud-processing capacity only as it is actually needed.*

The screenshot shows the Confluent Flink console interface. The left sidebar contains navigation links for Home, Environments, prod, Environment overview, Clusters, Flink, Streaming agents, Network management, Stream Governance, Schema Registry, Catalog management, and Integrations. The main panel displays the 'Flink' section with tabs for Statements, Materialized tables, Compute pools, API keys, Artifacts, and Endpoints. A search bar and filters are visible at the top of the table. The table lists 14 statements, showing their status, type, CFU usage, state size, creation time, and message backlog.

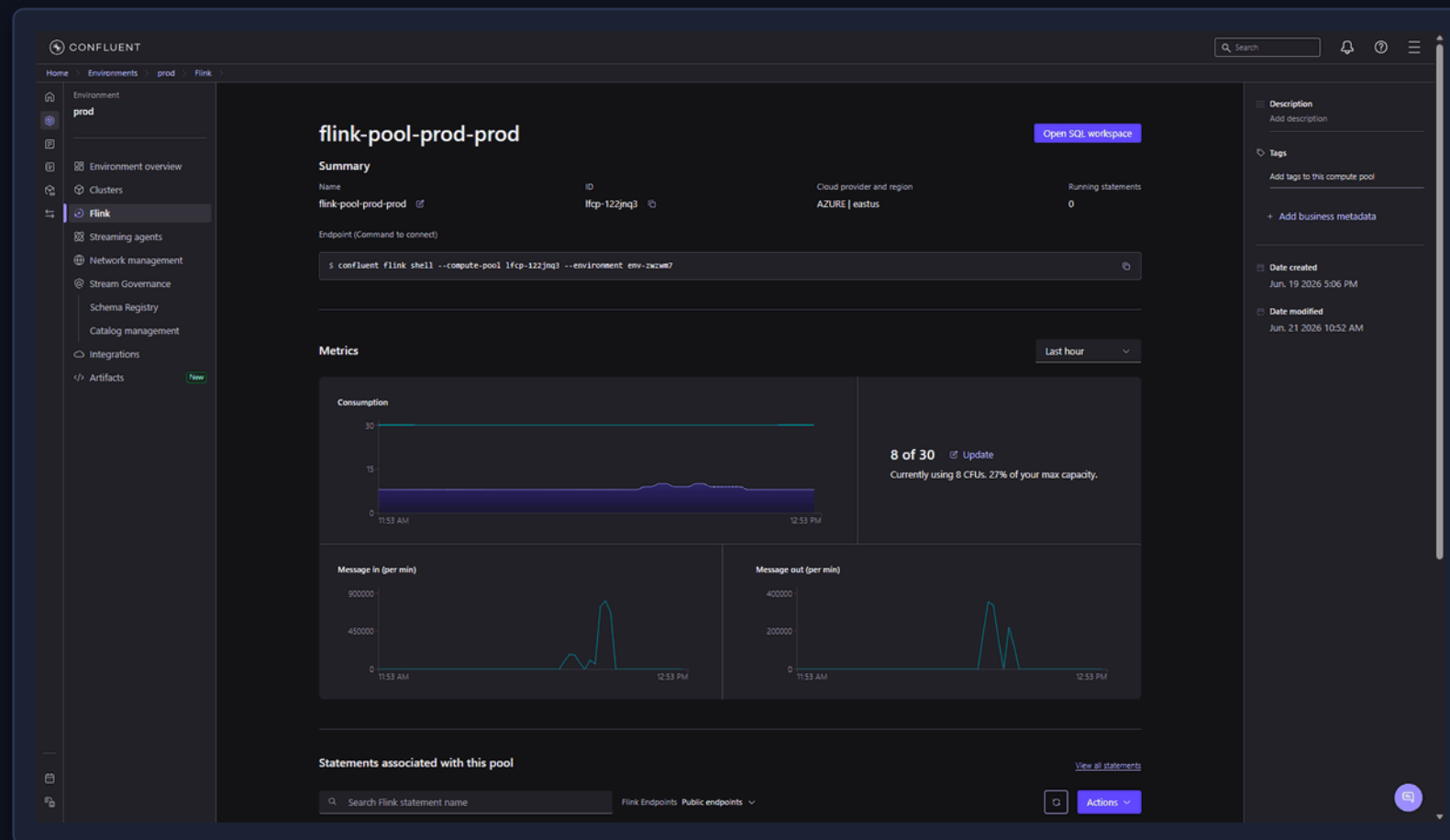
Flink statement name	Status	Statement type	Statement CFU	State size GB	Created	# Messages behind total
stmt-decide flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	0	18 hours ago	0 Not behind
stmt-complaint flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	0	18 hours ago	0 Not behind
stmt-enrich flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	0	19 hours ago	0 Not behind
stmt-ingest flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	0	19 hours ago	0 Not behind
stmt-rules flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	0	19 hours ago	0 Not behind
stmt-publish flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	0	19 hours ago	0 Not behind
stmt-initel flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	0	19 hours ago	0 Not behind
ad-hoc-describe-2790232 flink-pool-prod-prod	Stopped	SELECT	-	-	20 hours ago	-
ad-hoc-describe-2789680 flink-pool-prod-prod	Stopped	SELECT	-	-	a day ago	-
smoke-27896559527-vali flink-pool-prod-prod	Stopped	SELECT	-	-	a day ago	-

Governance is built into the platform

WHAT YOU'RE SEEING

- Stream Governance package: Advanced
- 99.99% uptime SLA, 20,000 contracts included
- 61 active data contracts registered today
- Schema Registry + Catalog for discovery and trust

WHY IT MATTERS *Trust and discoverability ship with the streams — not bolted on later.*



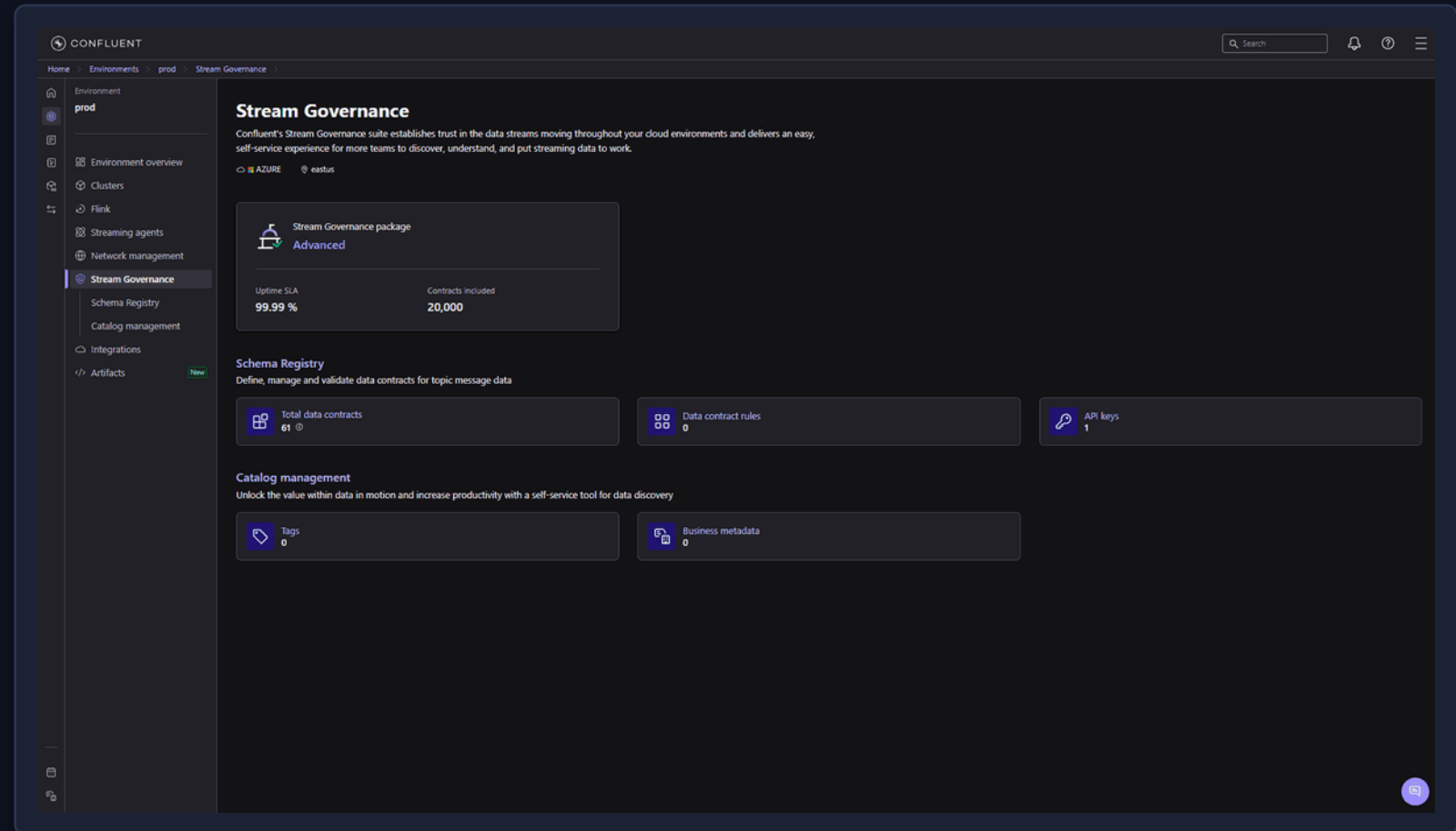
GOVERNANCE

Advanced tier: quality, catalog and lineage

WHAT YOU'RE SEEING

- Schema validation plus data-contract rules
- Catalog: tags, business metadata, REST & GraphQL APIs
- Real-time and point-in-time stream lineage
- All of it for \$1 per hour

WHY IT MATTERS *Enterprise data governance without standing up extra tooling.*



Every topic is an Avro contract

WHAT YOU'RE SEEING

- Subjects registered for both keys and values
- BACKWARD compatibility enforced on every evolution
- Lookup, alert and intelligence schemas all present
- 'View topics' links each contract to its stream

WHY IT MATTERS *No producer can publish a change that silently breaks a consumer.*

The screenshot displays the Confluent Stream Governance interface. The left sidebar shows the navigation menu with 'Stream Governance' selected. The main content area is titled 'Stream Governance' and shows the 'Advanced' package details. A modal window titled 'Stream Governance package' is open, displaying the following information:

- Stream Quality:**
 - Schema Registry and validation: 99.99 %
 - Uptime SLA: 20,000
 - Contracts included: Read: 75 | Write: 25
 - Schema Registry calls per second: 25
 - Data contract rules: 1
- Stream Catalog:**
 - Data organization and discoverability: 1
 - Automated ingestion of technical data: 1
 - Tags metadata: 1
 - UI and Rest API: 1
 - Business metadata: 1
 - GraphQL API: 1
- Stream Lineage:**
 - Data origin and tracking: 1
 - Real-time data streams lineage: 1
 - Graphical UI of event streams and data relationships: 1
 - Point-in-time lineage: 1
 - Lineage search: 1

The modal also shows a price of '\$1 per hour' and links for 'Questions? Contact customer support' and 'View more packages details'.

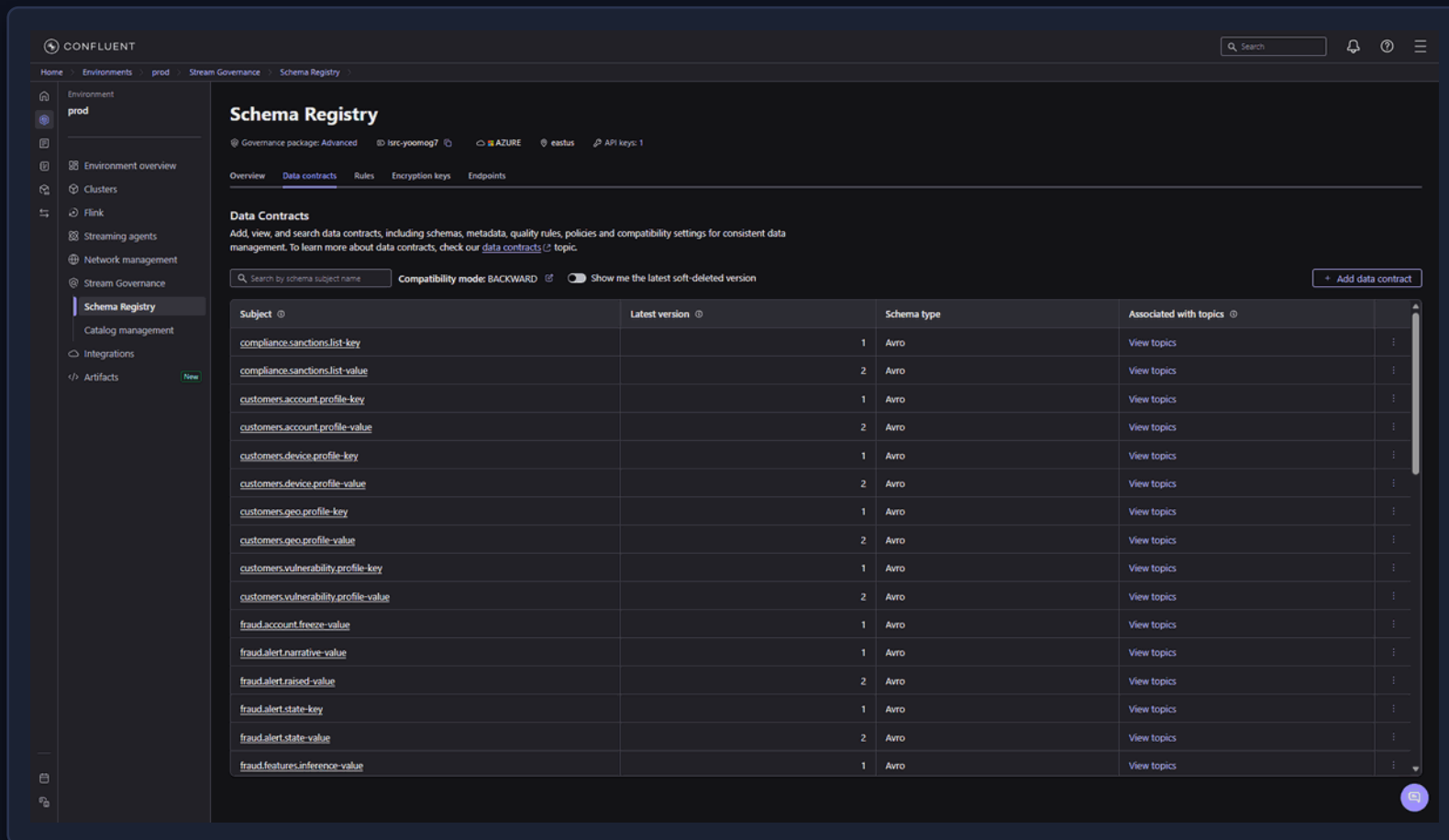
SCHEMA

A shared case-open contract for fraud & AML

WHAT YOU'RE SEEING

- One envelope feeds payments.case.fraud and .aml
- case_type enum is the FRAUD / AML discriminator
- Produced by STMT-DECIDE on BLOCK / HARD_BLOCK
- Versioned, Backward-compatible and governed

WHY IT MATTERS Schemas are the locked contract between Flink and the agent layer.



The screenshot displays the Confluent Schema Registry web interface. The left sidebar shows the navigation menu with 'Schema Registry' selected. The main panel is titled 'Schema Registry' and shows the 'Data contracts' tab. A table lists various data contracts, including compliance sanctions, customer profiles, and fraud alerts, each with a subject, latest version, schema type (Avro), and a link to view topics.

Subject	Latest version	Schema type	Associated with topics
compliance.sanctions.list-key	1	Avro	View topics
compliance.sanctions.list-value	2	Avro	View topics
customers.account.profile-key	1	Avro	View topics
customers.account.profile-value	2	Avro	View topics
customers.device.profile-key	1	Avro	View topics
customers.device.profile-value	2	Avro	View topics
customers.geo.profile-key	1	Avro	View topics
customers.geo.profile-value	2	Avro	View topics
customers.vulnerability.profile-key	1	Avro	View topics
customers.vulnerability.profile-value	2	Avro	View topics
fraud.account.freeze-value	1	Avro	View topics
fraud.alert.narrative-value	1	Avro	View topics
fraud.alert.raised-value	2	Avro	View topics
fraud.alert.state-key	1	Avro	View topics
fraud.alert.state-value	2	Avro	View topics
fraud.features.inference-value	1	Avro	View topics

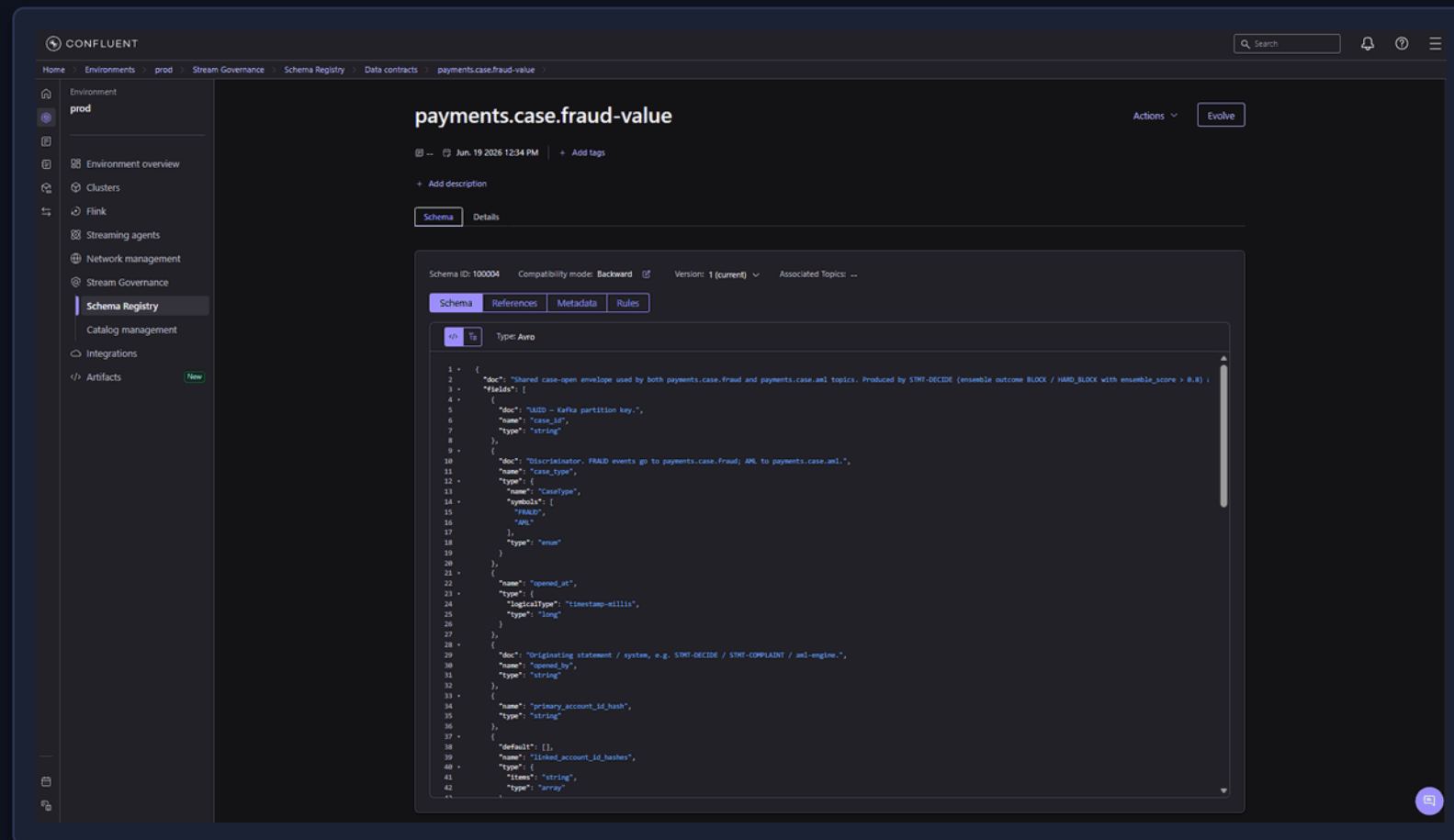
SCHEMA

The enriched event carries the full context

WHAT YOU'RE SEEING

- Output of STMT-ENRICH, version 3, Avro
- Self-contained: validated fields + 11 lookup joins
- Consumed by rules, publish, MuleHunter and AML
- Schemas register independently — no fragile references

WHY IT MATTERS *Downstream consumers get full context without re-querying a database.*

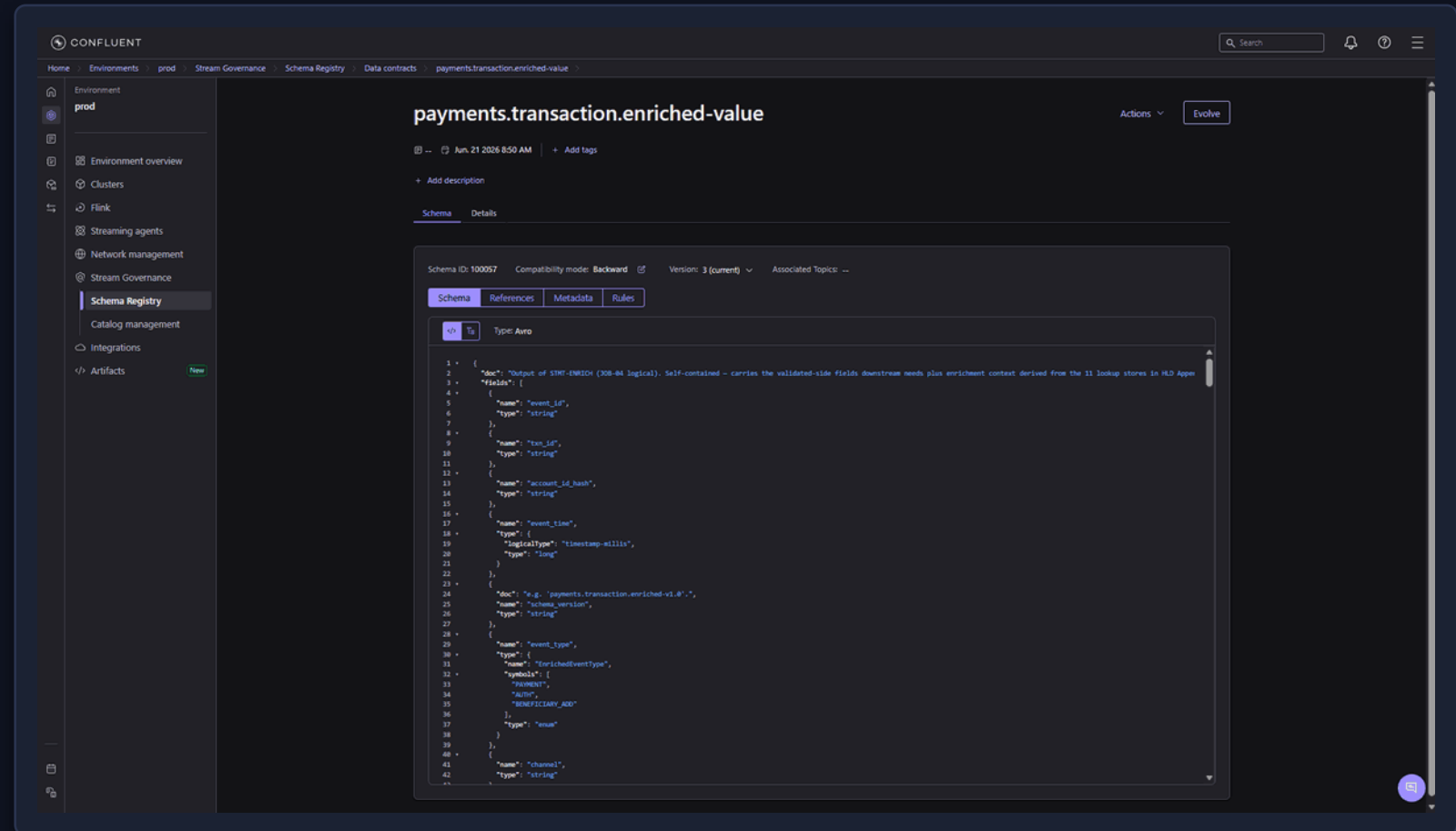


The entry point: every payment event

WHAT YOU'RE SEEING

- payments.transaction.raw, 24 partitions on Azure
- PaymentTransaction schema, 46 fields, PII pre-hashed
- Lineage shows producer → topic → consumers
- Iceberg / Tableflow export is one click away

WHY IT MATTERS *Raw events enter the platform already PII-safe and schema-validated.*



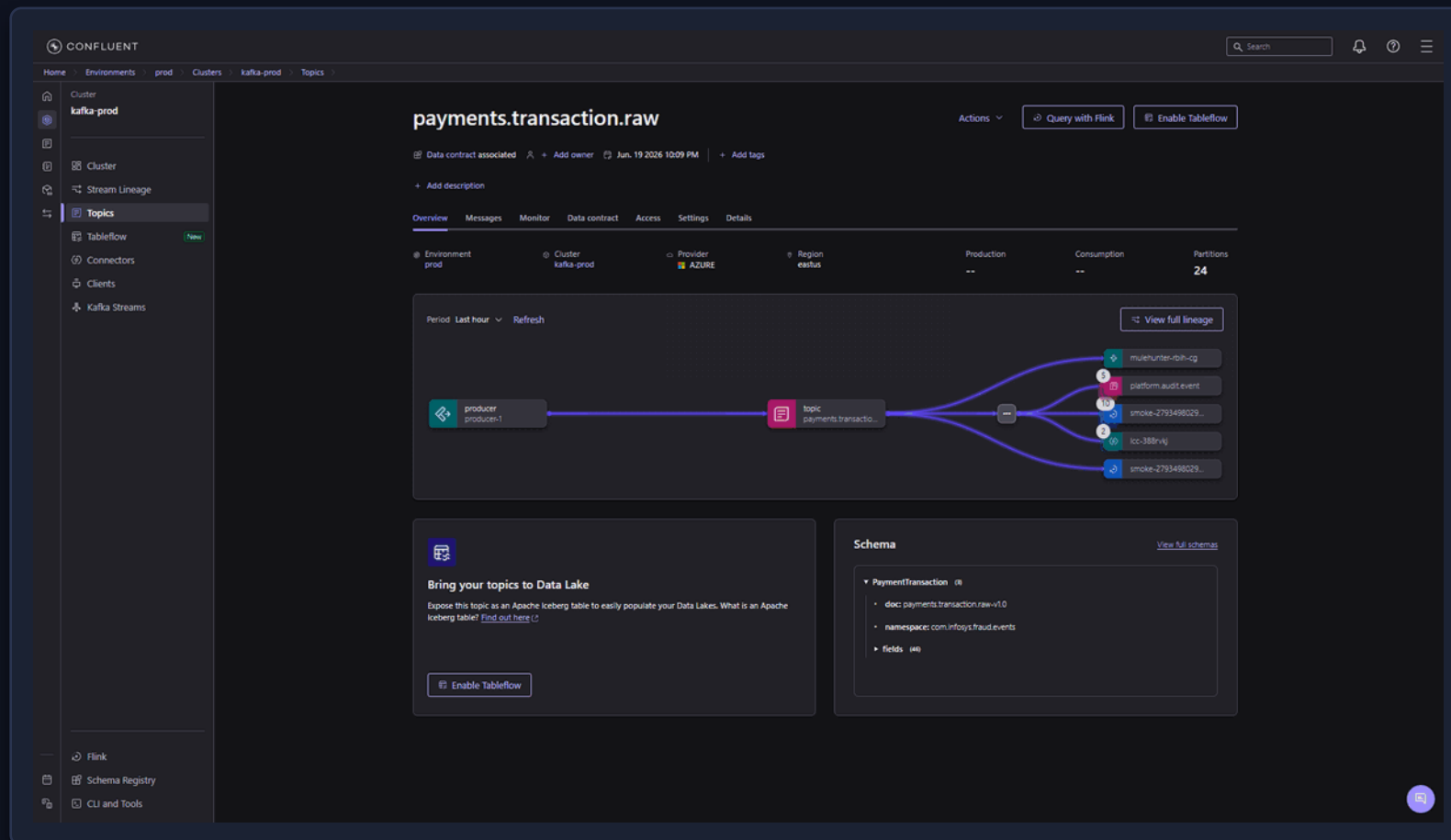
TOPICS

Enriched events are the hot fan-out

WHAT YOU'RE SEEING

- Output of STMT-ENRICH, 38 fields per event
- Read by STMT-RULES, STMT-PUBLISH, MuleHunter, AML
- Drives feature vectors to fraud.features.inference
- Data contract associated; lineage live

WHY IT MATTERS *A single enriched stream powers rules, ML scoring and analytics alike.*

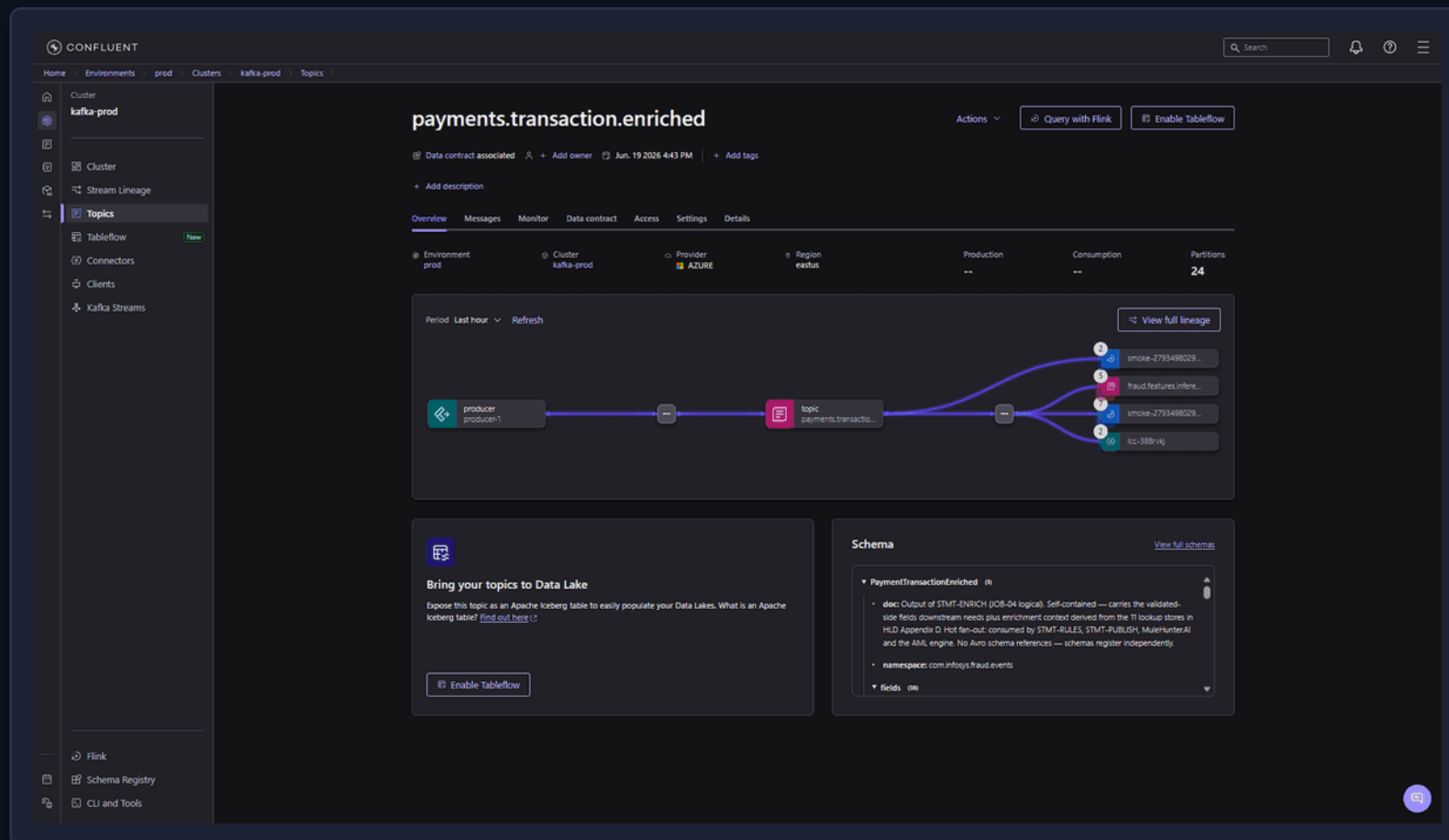


Validated events: the replay anchor

WHAT YOU'RE SEEING

- Output of STMT-INGEST: dedup + rail classify + sanity
- Carries forward every field STMT-ENRICH needs
- Independent Avro types — no schema references
- A clean checkpoint between ingest and enrichment

WHY IT MATTERS *A deterministic, replayable boundary partway through the pipeline.*



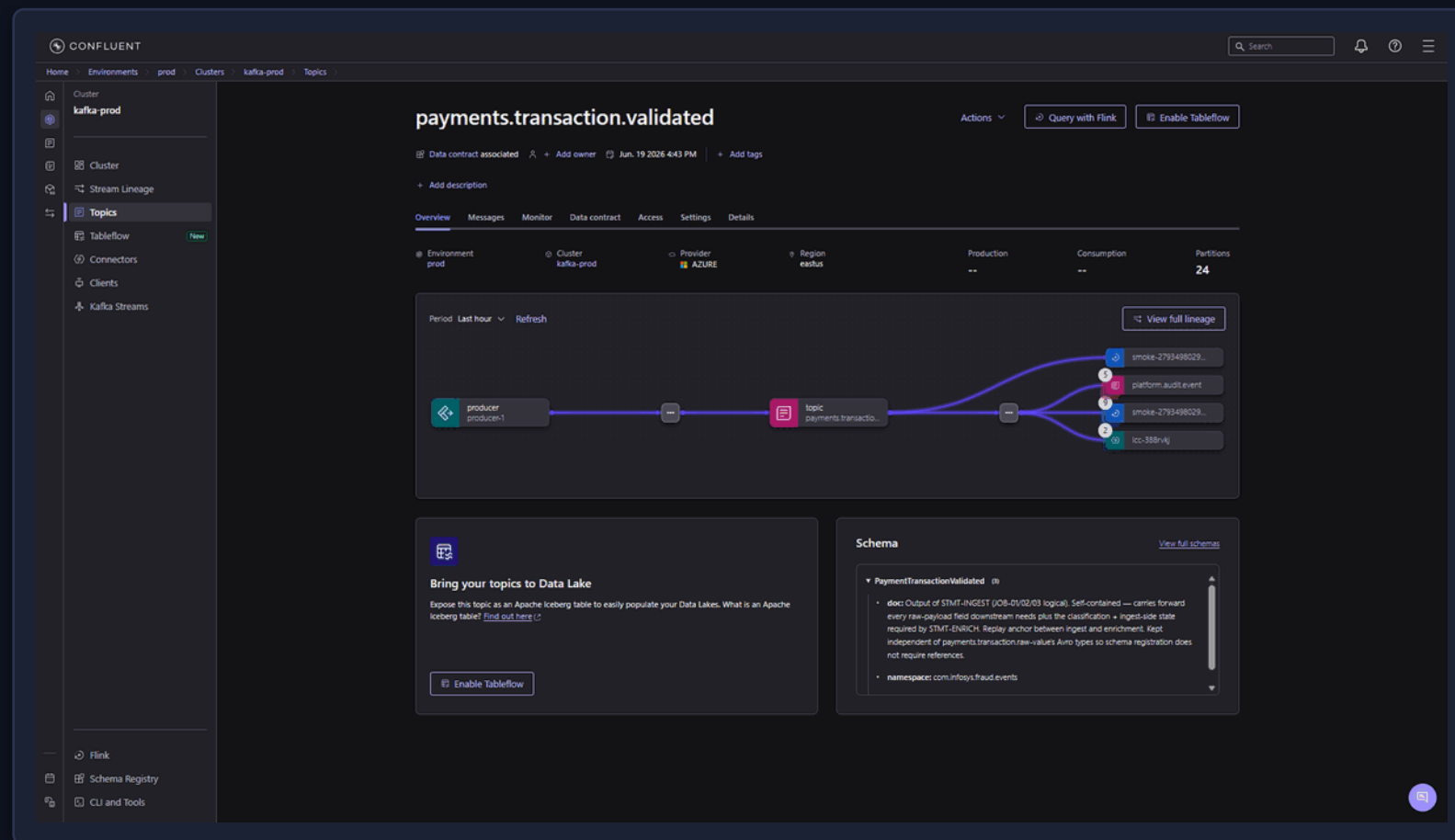
COST

The full platform runs on free credits

WHAT YOU'RE SEEING

- June 2026 accrued charges: \$0.00
- ~\$560 of usage fully offset by program promos
- Remaining free credits tracked in-console
- Transparent, itemised billing

WHY IT MATTERS *Production-grade architecture demonstrated at effectively zero cost.*



COST

Where the (offset) spend actually goes

WHAT YOU'RE SEEING

- Flink CFUs, Kafka CKUs and Connect tasks itemised
- Three Postgres connectors + the CDC source visible
- Advanced Stream Governance billed hourly
- Private-link network and storage broken out

WHY IT MATTERS *Clear per-resource cost visibility for real capacity planning.*

The screenshot displays the Confluent Billing and payment interface. The left sidebar contains navigation links: Home, Environments, Data Portal, SQL workspaces, Cluster links, and Migration hub (marked as New). The main content area is titled 'Invoice Breakdown' and includes a 'Select invoice range' section with dropdowns for Contract (All), Month (June), and Year (2026). Below this, the current accrued charges are \$0.00 USD for the period Jun. 1 - Jun. 30, 2026, with free credits of \$65 / \$75 USD remaining. A table lists various usage items and their associated charges, including usage charges and several promotional credits. The total accrued charges are \$0.00. A 'Download invoice' button is located below the table. The 'Invoice period details' section shows charges by environment (environment - prod \$518.66 USD). The 'Non-cluster-specific charges' section is partially visible at the bottom.

Resource name	Quantity	Unit price	Total
Usage			\$560.51
Promo applied (CONFLUENTDEV1)			-\$1.00
Promo applied (FLINKJAVA101)			-\$9.51
Promo applied (101CONNECT)			-\$25.00
Promo applied (KAFKA101)			-\$25.00
Promo applied (SCHEMA101)			-\$25.00
Promo applied (101SECURITY)			-\$25.00
Promo applied (DOTNETKAFKA101)			-\$25.00
Promo applied (FLINKSQL25)			-\$25.00
Promo applied (FREETRIAL400)			-\$400.00
Total accrued charges			\$0.00

AI narration runs inside Flink — live

WHAT YOU'RE SEEING

- CREATE MODEL gemini_narrate (Google Gemini backend)
- ML_PREDICT invoked straight from Flink SQL
- A HOLD alert → a plain-English analyst note in seconds
- Empathetic, PII-free and stamped with the model id

WHY IT MATTERS *Post-decision AI explains the alert — it never makes the decision (INV-3).*

The screenshot shows the Confluent Cloud interface with a sidebar on the left containing links to Home, Environments, Data Portal, SQL workspaces, Cluster links, and Migration hub. The main content area is titled 'Billing and payment' and displays three tables of charges.

Non-cluster-specific charges

Resource name	Quantity	Unit price	Total
flink-pool-prod-prod (flcp-122jmq3)	FlinkNumCPUs: 521.6333 CPU-hour	\$0.21 / CPU-hour	\$109.54
kafka-prod (lkc-pggr9j5)	ConnectCapacity: 51.0000 Hour	\$0.27777778 / Hour	\$14.17
Always On Stream Governance Package (lrc-yoomog7)	GovernanceBase: 97.0000 Hour	\$1 / Hour	\$97.00

Kafka cluster ID: lkc-mvv8vk1

Resource name	Quantity	Unit price	Total
kafka-prod (lkc-mvv8vk1)	KafkaNumCPUs: 8.0000 CPU-hour	\$2.25 / CPU-hour	\$18.00

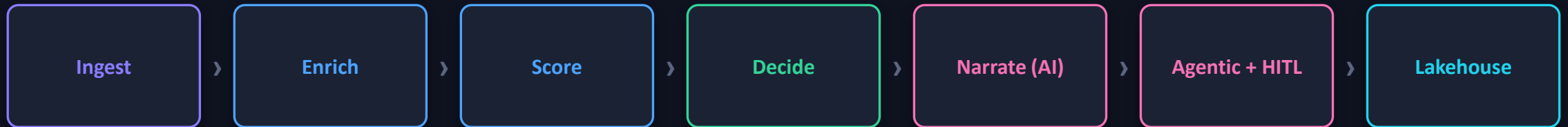
Kafka cluster ID: lkc-pggr9j5

Resource name	Quantity	Unit price	Total
postgres-sink-alerts (lcc-388rvk)	ConnectNumTasks: 46.0000 Task-hour	\$0.13416667 / Task-h...	\$6.17
postgres-sink-alerts (lcc-388rvk)	ConnectNumTasks: 3.0000 Task-hour	\$0.10416667 / Task-h...	\$0.31
postgres-sink-ops (lcc-6kko3g3)	ConnectNumTasks: 1.0000 Task-hour	\$0.13416667 / Task-h...	\$0.13
postgres-sink-ops (lcc-6kko3g3)	ConnectNumTasks: 2.0000 Task-hour	\$0.13416667 / Task-h...	\$0.27
postgres-sink-ops (lcc-7jyne1j)	ConnectNumTasks: 3.0000 Task-hour	\$0.10416667 / Task-h...	\$0.31
postgres-sink-ops (lcc-7jyne1j)	ConnectNumTasks: 4.0000 Task-hour	\$0.13416667 / Task-h...	\$0.54
postgres-sink-ops (lcc-mvv6867)	ConnectNumTasks: 41.0000 Task-hour	\$0.13416667 / Task-h...	\$5.50
postgres-sink-ops (lcc-mvv6867)	ConnectNumTasks: 1.0000 Task-hour	\$0.13416667 / Task-h...	\$0.13
postgres-cdc-source-lookups (lcc-mvnm7v)	ConnectNumTasks: 1.0000 Task-hour	\$0.53 / Task-hour	\$0.53
postgres-sink-ops (lcc-mvvoqd)	ConnectNumTasks: 2.0000 Task-hour	\$0.13416667 / Task-h...	\$0.27
postgres-sink-ops (lcc-mvvoqd)	ConnectNumTasks: 1.0000 Task-hour	\$0.13416667 / Task-h...	\$0.13
postgres-cdc-source-lookups (lcc-zmmg200)	ConnectNumTasks: 48.0000 Task-hour	\$0.53 / Task-hour	\$25.44
kafka-prod (lkc-pggr9j5)	KafkaNetworkWrite, private-link: 7.6485 GB	\$0.04 / GB	\$0.31
kafka-prod (lkc-pggr9j5)	KafkaNetworkRead, private-link: 11.6213 GB	\$0.04 / GB	\$0.46
kafka-prod (lkc-pggr9j5)	KafkaStorage: 370.5506 GB-hour	\$0.00009959 / GB-h...	\$0.04
kafka-prod (lkc-pggr9j5)	KafkaNumCPUs: 74.0000 CPU-hour	\$3.2351 / CPU-hour	\$239.40

At the bottom left, there are links to 'Schedule a help call' and 'File Support ticket'. At the bottom right, there is a small blue circle with the number '15'.

FROM RAW PAYMENT TO GOVERNED DECISION

One platform, end to end



PII-Safe

Identifiers hashed at source; raw PII never touches Kafka.

Deterministic Core

Rules, CEP and a 40/40/20 ensemble decide — replayable.

Governed AI

Gateway + Flink connections; agents assist, never decide.

IaC + Open Lakehouse

Terraform everything; Tableflow → Iceberg, zero ETL.

Confluent Kafka + Flink · Schema Registry · Tableflow → Apache Iceberg · HashiCorp Terraform · Governed Agentic AI · on Microsoft Azure